

Trust in scientists and their role in society across 68 countries

Viktoria Cologna^{1,2*}, Niels G. Mede², Sebastian Berger³, John Besley⁴, Cameron Brick^{5,6}, Marina Joubert⁷, Edward W. Maibach⁸, Sabina Mihelj⁹, Naomi Oreskes¹, Mike S. Schäfer², Sander van der Linden¹⁰, Nor Izzatina Abdul Aziz¹¹, Suleiman Abdulsalam¹², Nurulaini Abu Shamsi¹³, Balazs Aczel¹⁴, Indro Adinugroho^{15,16}, Eleonora Alabrese¹⁷, Alaa Aldoh⁵, Mark Alfano¹⁸, Innocent Mbuli Ali¹⁹, Mohammed Alsobay²⁰, Marlene Altenmüller²¹, R. Michael Alvarez²², Richard Amoako²³, Tabitha Amollo²⁴, Patrick Ansah²³, Denisa Apriliawati²⁵, Flavio Azevedo²⁶, Ani Bajrami²⁷, Ronita Bardhan²⁸, Keagile Bati²⁹, Eri Bertsou³⁰, Cornelia Betsch³¹, Apurav Yash Bhatiya³², Rahul Bhui^{20,33}, Olga Białobrzaska³⁴, Michał Bilewicz³⁵, Ayoub Bouguettaya³⁶, Katherine Breeden³⁷, Amélie Bret³⁸, Ondrej Buchel³⁹, Pablo Cabrera-Álvarez⁴⁰, Federica Cagnoli⁴¹, André Calero Valdez⁴², Timothy Callaghan⁴³, Rizza Kaye Cases⁴⁴, Sami Çoksan⁴⁵, Gabriela Czarnek⁴⁶, Steven De Peuter⁴⁷, Ramit Debnath^{22,48}, Sylvain Delouvée⁴⁹, Celia Díaz-Catalán⁴⁰, Lucia Di Stefano⁴¹, Kimberly C. Doell⁵⁰, Simone Dohle⁵¹, Karen M. Douglas⁵², Charlotte Dries⁵³, Dmitrii Dubrov⁵⁴, Małgorzata Dzimińska⁵⁵, Ullrich K. H. Ecker⁵⁶, Christian T. Elbaek⁵⁷, Mahmoud Elsherif⁵⁶, Benjamin Enke⁵⁸, Tom W. Etienne⁵⁹, Matthew Facciani⁶⁰, Antoinette Fage-Butler⁶¹, Md. Zaki Faisal⁶², Xiaoli Fan⁶³, Christina Farhart⁶⁴, Christoph Feldhaus⁶⁵, Marinus Ferreira¹⁸, Stefan Feuerriegel⁶⁶, Helen Fischer^{67,68}, Jana Freundt⁶⁹, Malte Friese⁷⁰, Simon Fuglsang⁷¹, Albina Gallyamova⁵⁴, Patricia Garrido-Vásquez⁷², Mauricio Garrido Vásquez⁷², Winfred Gatua⁷³, Oliver Genschow⁷⁴, Omid Ghasemi⁷⁵, Theofilos Gkinopoulos⁴⁶, Jamie Gloor⁷⁶, Ellen Goddard⁶³, Mario Gollwitzer²¹, Claudia González Brambila⁷⁷, Hazel Gordon¹⁵, Dmitry Grigoryev⁵⁴, Gina M. Grimshaw⁷⁸, Lars Guenther⁷⁹, Håvard Haarstad⁸⁰, Dana Harari⁸¹, Lelia N. Hawkins⁸², Przemysław Hensel⁸³, Alma Cristal Hernández-Mondragón⁸⁴, Atar Herziger⁸¹, Guanxiong Huang⁸⁵, Markus Huff⁶⁷, Mairéad Hurley⁸⁶, Nygmet Ibadildin⁸⁷, Maho Ishibashi⁸⁸, Mohammad Tarikul Islam⁸⁹, Younes Jeddi¹², Tao Jin⁹⁰, Charlotte A. Jones⁹¹, Sebastian Jungkunz⁹², Dominika Jurgiel⁹³, Zhangir Kabdulkair⁸⁷, Jo-Ju Kao⁹⁴, Sarah Kavassalis⁸², John R. Kerr⁹⁵, Mariana Kitsa⁹⁶, Tereza Klabíková Rábová⁹⁷, Olivier Klein⁹⁸, Hoyoun Koh⁹⁹, Aki Koivula¹⁰⁰, Lilian Kojan⁴², Elizaveta Komyaginskaya⁵⁴, Laura König¹⁰¹, Lina Koppel¹⁰², Alexandra Kosachenko¹⁰³, John Kotcher⁸, Laura S. Kranz⁷⁸, Pradeep Krishnan³⁰, Silje Kristiansen^{104,105}, André Krouwel¹⁰⁶, Toon Kuppens¹⁰⁷, Eleni A. Kyza¹⁰⁸, Claus Lamm⁵⁰, Anthony Lantian¹⁰⁹, Aleksandra Lazić¹¹⁰, Oscar Lecuona de la Cruz¹¹¹, Jean-Baptiste Légal¹⁰⁹, Zoe Leviston¹¹², Neil Levy^{18, 113}, Amanda Lindkvist¹⁰², Grégoire Lits¹¹⁴, Andreas Löschel⁶⁵, Alberto López Ortega¹⁰⁶, Carlos Lopez-Villavicencio¹¹⁵, Nigel Mantou Lou¹¹⁶, Chloe H. Lucas⁹¹, Kristin Lunz-Trujillo^{117,118}, Mathew D. Marques¹¹⁹, Sabrina Mayer⁹², Ryan McKay¹²⁰, Hugo Mercier¹²¹, Julia Metag¹²², Taciano L. Milfont¹²³, Joanne Miller¹²⁴, Panagiotis Mitkidis⁵⁷, Fredy Monge-Rodríguez¹¹⁵, Matt Motta⁴³, Iryna Mudra⁹⁶, Zarja Muršič¹²⁵, Jennifer Namutebi¹²⁶, Eryn J. Newman¹¹², Jonas P. Nitschke⁵⁰, Fernando Luiz Nobre Cavalcante¹²⁷, Ntui-Njock Vincent Ntui¹²⁸, Daniel Nwogwugwu¹²⁹, Thomas Ostermann¹³⁰, Tobias Otterbring¹³¹, Jaime Palmer-Hague¹³², Myrto Pantazi⁹⁸, Philip Pärnamets¹³³, Paolo Parra Saiani⁴¹, Mariola Paruzel-Czachura^{134,135}, Michal Parzuchowski³⁴, Yuri Pavlov¹⁰³, Adam R. Pearson¹³⁶, Myron A. Penner¹³², Charlotte R. Pennington¹³⁷, Katerina Petkanopoulou¹³⁸, Marija Petrović¹¹⁰, Jan Pfänder¹²², Dinara Pisareva⁹⁹, Adam Ploszaj¹³⁹, Karolína Poliaková⁹⁷, Ekaterina Pronizius⁵⁰, Katarzyna Pypno¹³⁴, Diwa Malaya Quiñones¹⁴⁰, Pekka Räsänen¹⁰⁰, Adrian Rauchfleisch⁹⁴, Felix G. Rebitschek^{53,141}, Cintia Refojo Seronero⁴⁰, Gabriel Rêgo¹⁴², James P. Reynolds¹³⁷, Joseph Roche⁸⁶, Simone Rödder¹⁴³, Jan Philipp Röer¹³⁰, Robert M. Ross¹⁸, Isabelle Ruin¹⁴⁴, Osvaldo Santos¹⁴⁵, Ricardo R. Santos¹⁴⁵, Philipp Schmid^{31, 146, 147}, Stefan Schulreich^{148,149}, Bermond Scoggins¹⁵⁰, Amena Sharaf¹⁵¹, Justin Sheria Nfundiko^{152, 153}, Emily Shuckburgh⁴⁸, Johan Six¹⁵⁴, Nevin Solak¹⁵¹, Leonhard Späth¹⁵⁴, Bram Spruyt¹⁵⁵, Olivier Standaert¹¹⁴, Samantha K. Stanley^{112, 156}, Gert Storms⁴⁷, Noel Strahm³, Stylianos Syropoulos¹⁵⁷, Barnabas Szaszi¹⁴, Ewa Szumowska⁴⁶, Mikihito Tanaka¹⁵⁸, Claudia Teran-Escobar¹⁴⁴, Boryana Todorova⁵⁰, Abdoul Kafid Toko¹², Renata Tokri¹⁵⁹, Daniel Toribio-Florez⁵², Manos Tsakiris^{120, 160}, Michael Tyralla¹⁶¹, Özden Melis Ulug¹⁶², Ijeoma Chinwe Uzoma¹⁶³, Jochem van Noord²⁶, Christiana Varda^{108,164}, Steven Verheyen¹⁶⁵, Iris Vilares⁹⁰, Madalina Vlasceanu¹⁶⁶, Andreas von Bubnoff¹⁶⁷, Iain Walker^{112, 168}, Izabela Warwas⁵⁵, Marcel Weber⁷⁰, Tim Weninger⁶⁰, Mareike Westfal⁷⁴, Florian Wintterlin¹²², Adrian Dominik Wojcik¹⁶⁹, Ziqian Xia¹⁷⁰, Jinliang Xie¹⁷¹, Ewa Zegler-Poleska¹³⁹, Amber Zenklusen³⁰, Rolf A. Zwaan¹⁶⁵

*Corresponding author. Email: viktoriacologna@gmail.com Author affiliations can be found on page 34.

Abstract

Science is crucial for evidence-based decision-making. Public trust in scientists can help decision-makers act based on the best available evidence, especially during crises^{1,2}. However, in recent years the epistemic authority of science has been challenged, causing concerns about low public trust in scientists³. We interrogated these concerns with a pre-registered 68-country survey of 71,922 respondents and find that in most countries, most people trust scientists and agree that scientists should engage more in society and policymaking. We find variations between and within countries, which we explain with individual- and country-level variables, including political orientation. While there is no widespread lack of trust in scientists, we cannot discount the concern that lack of trust in scientists by even a small minority may affect considerations of scientific evidence in policymaking. These findings have implications for scientists and policymakers seeking to maintain and increase trust in scientists.

Main text

Public trust in science provides many benefits to society. It helps people make informed decisions (e.g., on health and nutrition) based on the best available evidence, provides the foundation for evidence-based policymaking, and facilitates government spending on research. Trust in science and scientists enables management of global crises like the COVID-19 pandemic and climate change. Societies with high public trust in science and scientists dealt with the COVID-19 pandemic more effectively, as citizens were more likely to comply with non-pharmaceutical COVID-19 interventions² and had higher vaccine confidence¹. People with high trust in scientists are also more likely to engage in individual and collective action on climate change^{4,5}.

Studies find that, globally, most people trust science, and scientists are among the most trusted actors in society⁶⁻⁸. Despite these findings, there is a popular dominant narrative claiming there is a crisis of trust in science and scientists^{3,9}. This narrative well predates the COVID-19 pandemic and may alter people's views about scientists¹⁰⁻¹². Therefore, it is important to revisit this narrative and provide robust empirical evidence on whether it is accurate.

Most previous studies have been limited to the Global North, typically the United States or Europe, including our own previous work (see e.g.,¹³⁻¹⁹). A few studies go beyond these regions^{6-8,20-23}. However, they assess a limited range of theoretical constructs. We address this limitation in two ways: First, we analyse the extent to which people believe that scientists should be involved in society and policymaking. We refer to this as *normative perceptions of science in society and policymaking*. Second, we investigate which issues people want scientists to prioritise in their work, and how such perceptions are related to their trust in scientists. Previous studies have shown that trust is affected by the perception of value alignment²⁴. People who feel that their concerns and values are not reflected in the priorities of scientists may therefore doubt the trustworthiness of scientists.

Our large-scale, pre-registered survey expands and strengthens previous work by offering the largest global dataset on trust in scientists post-COVID-19 pandemic¹⁶, and by investigating the public's normative perceptions of the role of scientists in society and policymaking and their desired research priorities. We use a theoretically informed multidimensional trust measure²⁵ and examine relevant demographic, ideological, attitudinal, and country-level factors to explain trust across countries⁷. We survey underrepresented countries and individuals in research²⁶ and, in almost all countries, have worked with local research partners^{27,28}.

Our study answers the following questions: 1) How much do people around the world trust scientists, and how do levels of trust vary across countries? 2) How do demographic, ideological, attitudinal, and country-level factors relate to trust in scientists (see fig. S1a for a directed acyclic graph), and how do these relationships vary between countries? 3) What are people's normative perceptions of scientists in society and policymaking, and how do they differ across countries? 4) What issues do people want scientists to prioritise, and do they believe that scientists actually address these priorities? See the preregistration for more detailed research questions and hypotheses: **[URL REDACTED]**

By investigating trust in scientists, we do not mean to imply that trust is always warranted. In some situations, low trust may be warranted. For example, science's fraught historical relationship with racism, its role in perpetuating racialized forms of knowledge production, sustaining racial paradigms²⁹, and disregarding ethical canons by experimenting on non-White human subjects³⁰, has reduced research participation in some populations³¹. Further, the epistemic authority of science and scientists has been challenged by mis- and disinformation^{32,33}, a "reproducibility crisis"³⁴, conspiracy theories^{35,36}, and science-related populist attitudes^{37,38}. Science-related populism has been conceptualised as a perceived antagonism between 'the ordinary people' and common sense versus academic elites and scientific expertise³⁷. Unlike political populism, which criticises political elites and their political power claims, science-related populism criticises academic elites, challenges their decision-making authority in scientific research, and suggests that their epistemic truth claims are inferior to the common sense of 'the people'³⁷. Anti-science attitudes, even if held by only a minority of people, raise concerns about a potential crisis of trust in science which could challenge the epistemic authority of science and the role of scientists in supporting evidence-based policymaking^{20,37}. These concerns, which have been prominently discussed in leading news media, have been exacerbated as trust in scientists and their desired role in policymaking has become divided along partisan lines. Several studies show that in the United States and some other countries, conservatives and right-leaning individuals have low levels of trust in scientists, hold stronger anti-science attitudes, and express low confidence that scientists act in the best interest of the public, provide benefits to society, and apply reliable methods^{19–21,39,40}. Global empirical evidence is needed to determine how widespread such critical attitudes toward science are across countries and population groups.

Our survey goes beyond commonly studied correlates of trust in scientists in four important ways. First, we investigate how trust in scientists relates to science-related populist attitudes. Science-related populists deny that scientists are knowledgeable experts and believe that they do not act in the interest of the general public: two key aspects of trust in scientists³⁷. Second, we investigate whether trust in scientists is related to people's social dominance orientation (SDO), which has been defined as “the degree to which individuals desire and support group-based hierarchy and the domination of ‘inferior’ groups by ‘superior’ groups”⁴¹ (p. 48). Individuals high in SDO are arguably less likely to trust scientists as they perceive universities as hierarchy-attenuating social institutions⁴². Previous research supports this, showing that SDO is a predictor of low trust in scientists⁴³ and distrust in climate science^{19,43}. However, it is unknown how SDO relates to trust in scientists across many countries. Third, we investigate what goals people want scientists to prioritise in their work and how this relates to trust. Fourth, we investigate whether people perceive that their desired priorities are tackled by science.

To answer our research questions, we conducted a crowd-sourced Many Labs project with the same translated online questionnaire given to 71,922 respondents in $k = 68$ countriesⁱ on all inhabited continents (fig. S2). The survey covered 31% of the world's countries that together make up 79% of the global population. Data were collected between November 2022 and August 2023, with quota samples that were weighted according to national distributions of age, gender, and education level, as well as country sample size. As recommended by other global studies on trust in scientists⁷, we provided respondents with a definition of science and scientists to mitigate semantic variations across languages (see SI)ⁱⁱ. We measured trust in scientists (instead of science) because “science” is more abstract than “scientists” and therefore makes a less clear referent: People may think of scientific institutions, scientific communities, scientific methods, or individual scientists when being asked about their general perception of “science”. However, these trust measures can be distinguished both conceptually and empirically^{25,44}. For example, research has shown that less educated people trust scientific *methods* more than scientific *institutions*⁴⁴. General measures that assess trust in the scientific community only capture some of the conceptually established dimensions of perceived trustworthiness (e.g., expertise)²⁵. We reduced this ambiguity by avoiding the abstract category “science,” and using the more concrete reference object “scientists”⁷. We slightly deviated from the preregistration because of multicollinearity (exclusion of confidence in science as a model covariate) and sparsely populated sample strata in certain countries (collapsing neighbouring strata during post-hoc weighting; see SI). All analyses can be reproduced with the replication materials available at: **[URL REDACTED]**

Trust in scientists across the world

We employed an index composed from a 12-item scale measuring four established dimensions of trustworthiness: perceived competence, benevolence, integrity, and openness^{13,25,45}. This scale is based on the most comprehensive review of trust measures used to assess trustworthiness perceptions of scientists⁴⁵. It was pre-tested to confirm its reliability, relies on accepted conceptual assumptions that we validated in factor analyses, and has high reliability across countries⁴⁶. However, confirmatory factor analyses show that we can only assume configural invariance and no metric or scalar invariance⁴⁶. This is a common caveat of multilingual survey research and to some extent unavoidable⁴⁷. When these components of trustworthiness perceptions are aggregated to a single score, the index represents an integrative measure of public trust in scientists with strong reliability (Cronbach's $\alpha = 0.93$ and $\omega = 0.95$). Therefore, we used the aggregate index for our main analyses (see SI for additional analyses with individual trust dimensions).

Globally, trust in scientists is *moderately* high (global $M = 3.62$, $SD = 0.70$; 1 = very low, 2 = somewhat low, 3 = neither high nor low, 4 = somewhat high, 5 = very high). No country shows low overall trust in scientists (Fig. 1). Across the globe, people perceive scientists as having high competence ($M = 4.02$, $SD = 0.71$), with 78% believing that scientists are qualified to conduct high-impact research (5% believe they are unqualified and 16% selected the scale midpoint; fig. S3). People perceive scientists to have moderate integrity ($M = 3.58$, $SD = 0.78$), and benevolent intentions ($M = 3.55$, $SD = 0.82$; table S1). For example, 57% of people believe that most scientists are honest (11% believe they are dishonest and 31% selected the scale midpoint) and 56% believe that most scientists are concerned about people's wellbeing (15% believe they are not concerned and 29% selected the scale midpoint). Scientists' perceived openness to feedback is slightly lower ($M = 3.33$, $SD = 0.86$), with 42% believing that scientists pay somewhat or very much attention to others' views. Globally, 75% agree that scientific research methods are the best way to find out if something is true or false. Trust in scientific methods moderately correlates with trust in scientists ($r = 0.47$, $p < .001$), supporting previous findings on the multidimensionality of trust in science⁴⁴.

While trust in science is moderately high overall, there are notable variations across countries and regions (Fig. 1). Contrary to previous studies^{7,8}, we do not find a clear pattern that scientists are less trusted in Latin American and African countries. However, we do find patterns within specific regions. For example, Russia as well as several former Soviet republics and satellite states, such as Kazakhstan, show relatively low trust in scientists.

Means and standard errors of trust in scientists across countries

Error bars show standard errors, vertical line indicates global mean

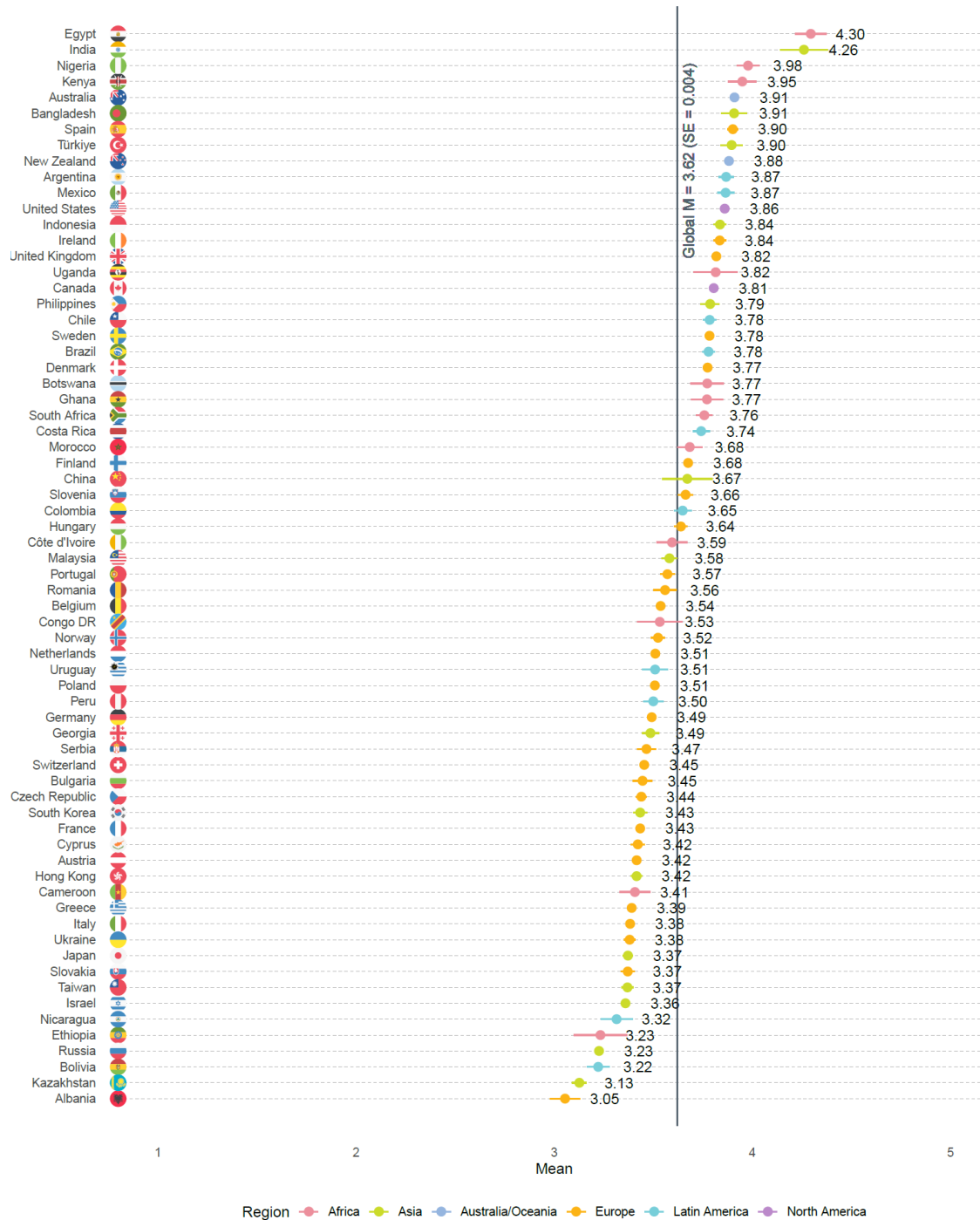
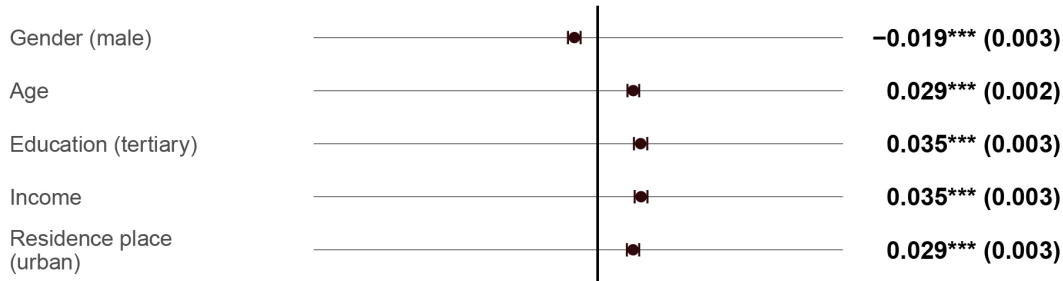


Fig. 1. Weighted means (M) for trust in scientists across countries and regions (1 = very low, 3 = neither high nor low, 5 = very high). Note. Vertical line denotes weighted global mean. Horizontal lines indicate standard errors (SE). Country-level SE s range between 0.008-0.133.

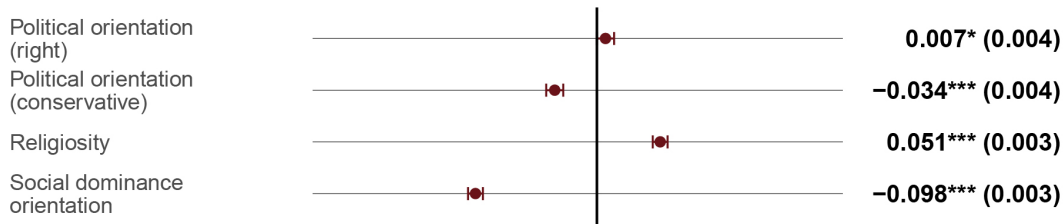
Correlates of trust in scientists and cross-country variations

To identify correlates of trust in scientists globally, we fitted linear random-intercept regression models that included post-stratification weights to provide estimates that are nationally representative in terms of gender, age, and education in almost all countries. To investigate how trust in scientists differs across population groups, we assessed several demographic variables and analysed their correlation with trust in scientists. We find higher levels of trust among many demographic groups: women, older people, residents of urban (versus rural) regions, people with high incomes, religious, educated, liberal, and left-leaning (Fig. 2, see also table S2). Differences across countries and sociodemographic groups can be explored with an online data visualisation tool developed for this project: **[URL REDACTED]**

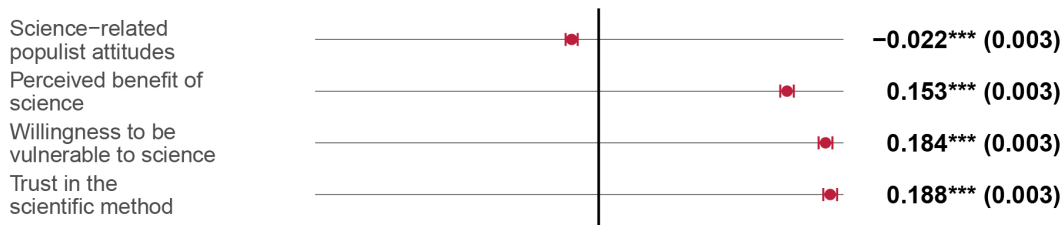
Block 1: Demographic characteristics



Block 2: Ideological views



Block 3: Attitudes to science



Block 4: Country-level indicators

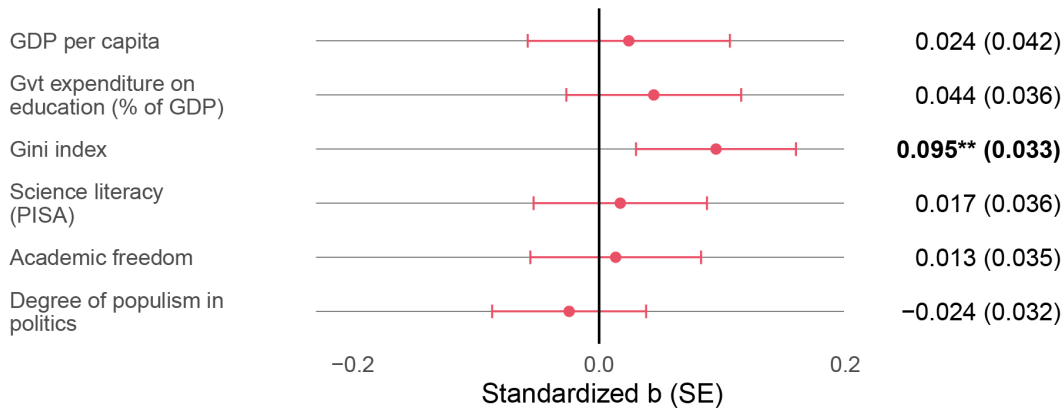


Fig. 2. Standardised estimates of weighted blockwise multilevel regression model testing the association of trust in scientists with demographic characteristics, attitudes, and country-level indicators (random intercepts across countries). Block 1 uses data from all 68 countries, block 2 uses data from 67 countries (all except Malaysia, where social dominance orientation was not measured), block 3 uses data from 66 countries (all except Malaysia and Mexico, where willingness to be vulnerable to science was not measured), block 4 uses data from 51 countries

(all except those where PISA's literacy scores were not available, see SI). Full regression results are reported in table S2. Results of exploratory analyses with individual trust dimensions are reported in fig. S4-S7. * $p < .05$, ** $p < .01$, *** $p < .001$.

One might assume that trust in science would correlate with tertiary education, as people with more years of schooling and university education have had more chances to build a closer relationship with science and experience the competence and benevolence of scientists, for example⁴⁸. However, our data shows only a small positive relationship between tertiary education and trust in scientists on average ($b = 0.035$, $p < .001$). In fact, in most countries we find little or no credible evidence for a relationship between tertiary education and trust (fig. S8). Overall, the relationships between demographic characteristics and trust in scientists are very small (marginal effects plots with unstandardized units are shown in fig. S9).

Many might also assume that religiosity is associated with lower trust in scientists, given that many studies conducted in Global North countries find this relationship (see e.g.,^{19,49}). However, against this assumption, one previous study found that only 29% of people worldwide believe that science stands in disagreement with their religion⁷. Another study found that while religiosity is associated with negative attitudes towards science in the United States, the relationship is inconsistent across the world⁵⁰. Indeed, we find that, globally, religiosity is positively associated with trust in scientists ($b = 0.051$, $p < .001$). However, as previous studies have also shown, we find substantive differences across countries and regions^{50,51}. In Muslim countries such as Türkiye, Bangladesh, and Malaysia (fig. S10), trust is positively associated with religiosity. In contrast, in many Christian-majority European countries, and in Israel, trust is negatively associated with religiosity. Qualitative interviews conducted by the Pew Research Center put these findings into context⁵². They found that most Muslim participants did not perceive a conflict between science and religion, because their holy text, the Quran, proclaims many principles of science. Conversely, some Christians perceive that science disagrees with their religion, even though there are pronounced variations across countries⁵². Our findings are consistent with these results.

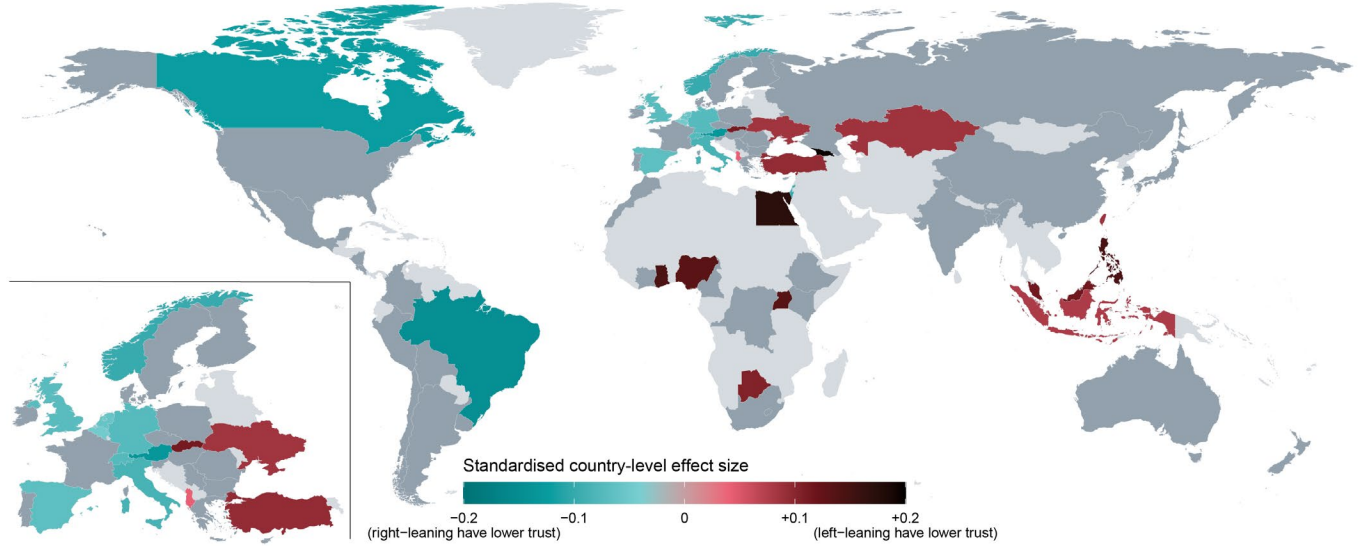
Other positive correlates of trust in scientists include people's willingness to rely on scientific advice and thus make themselves vulnerable to scientists, the belief that science benefits people like them, and trust in scientific methods.

Our study also sheds light on individual attributes that are associated with *lower* trust in scientists, namely conservative political orientation, social dominance orientation, and science-populist attitudes. Previous studies, which mostly focused on North America and Europe, have found right-leaning and conservative political orientation to be negatively associated with trust in scientists^{19,20}. Our study partly confirms these findings. We find a negative association between trust and conservative political orientation. However, we find a very small, positive relationship between right-leaning political orientation and trust. Given that some recent global social science

studies used a *left-right* measure to assess political orientation while others used a *liberal-conservative* measure^{53–55}, we used both measures and analysed how results vary depending on the measure in question. We find that the relationships between the two measures of political orientation and trust vary substantially across countries (Fig. 3a and 3b; fig. S11-S12). For example, in the United States, trust is associated with a liberal orientation but not with one's self-placement on the left-right spectrum. More generally, right-leaning and conservative political orientation are negatively associated with trust in scientists in several European and North American countries, so previous research, which has disproportionally focused on these countries, has tended to stress right-leaning and conservative distrust. However, in most countries ($k = 41$ for left-right measure and $k = 48$ for liberal-conservative measure), our data do not show credible evidence of a relationship between political orientation and trust in scientists. Furthermore, in some Eastern European, South-East Asian, and African countries, right-leaning individuals have higher trust in scientists. These contrasting findings may be explained by the fact that in some countries right-leaning parties may have cultivated reservations against scientists among their supporters, while in other countries left-leaning parties may have done so⁵⁶ (see fig. S11). In other words, the attitudes of political leadership rather than peoples' political orientation may better explain politically correlated attitudes towards scientists (see SI for selected country-specific explanations). We encourage future research to investigate differences in the two measures of political orientation on the country level (for broader discussions on these measures see^{57–60}).

Relationship of political orientation and trust in scientists

3a. Country-level effects of right-leaning political orientation and trust in scientists



3b. Country-level effects of conservative political orientation and trust in scientists

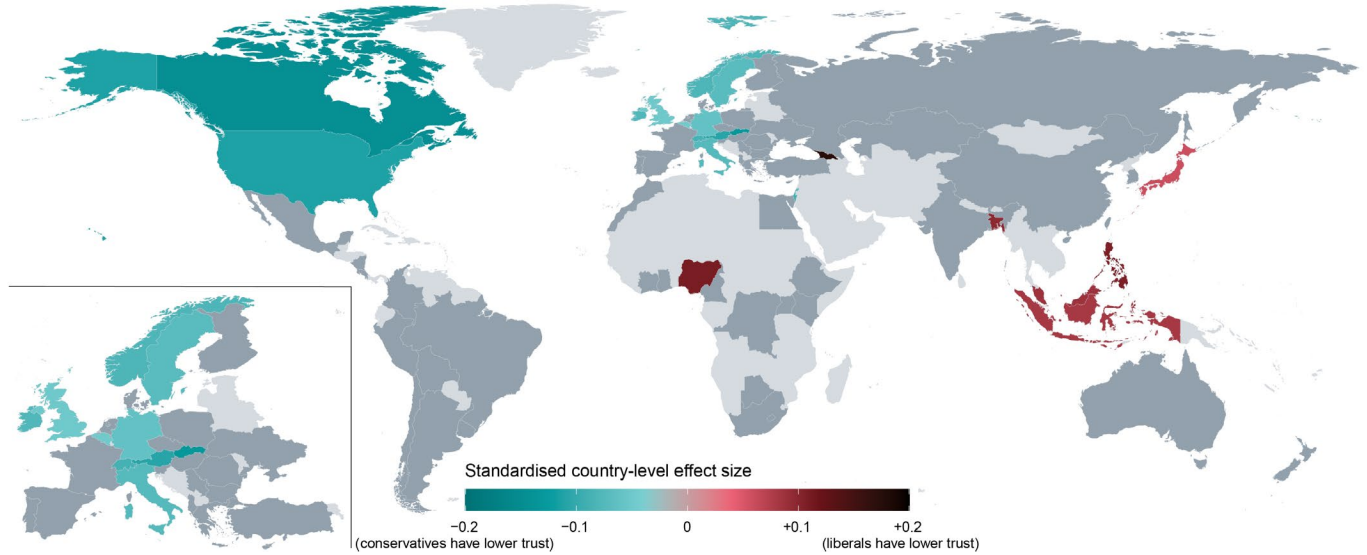


Fig. 3. Relationship of political orientation measures and trust in scientists. Visualises standardised country-level effects of political orientation (3a: 1 = left – 5 = right; 3b: 1 = liberal – 5 = conservative) on trust in scientists (1 = very low, 3 = neither high nor low, 5 = very high). These effects are sums of the global effect for political orientation and the random effect within each country; they were estimated with weighted linear multilevel regressions that contained random intercepts and random slopes for political orientation (3a: left-right; 3b: liberal-conservative) across countries. These models control for demographic characteristics. Significance tests used percentile bootstrapping. Countries with significant country-level effects ($p < .05$) are displayed in colours: Countries coloured in shades of blue show a positive country-

level association of left-leaning (3a) respectively liberal (3b) orientation and trust in scientists (i.e., right-leaning resp. conservatives have lower trust). Countries coloured in shades of red show a positive country-level association of right-leaning (3a) respectively conservative (3b) orientation and trust in scientists (i.e., left-leaning resp. liberals have lower trust). Countries with non-significant effects are shaded in dark grey. Countries with no available data are shaded in light grey.

Some studies have looked at social dominance orientation—that is to say, a preference for social hierarchy and inequality—and found it to be negatively associated with trust in scientists^{19,43}. Our results confirm this: The low global mean for SDO ($M = 3.62$, $SD = 1.76$; 1 = extremely oppose – 10 = extremely favor) is consistent with the overall moderately high trust in scientists. Moreover, we find that those who favour hierarchy enhancement (i.e., more strongly endorse SDO) are less likely to trust scientists ($b = -0.098$, $p < .001$). This may be because they see universities (i.e., scientists) as institutions that weaken social hierarchies⁴².

We also find that low trust in scientists is associated with science-related populist attitudes—that is, beliefs that people’s common sense is superior to the expertise of scientists and scientific institutions. This corroborates findings on single countries³⁸, and is first global evidence that populist resentment against science, a prevalent component of the trust crisis narrative, may undermine public trust in scientists.

We also tested pre-registered hypotheses assuming that trust in scientists is linked to country-level indicators, including GDP per capita, PISA’s science literacy score, and the Academic Freedom index. Contrary to the finding of the Wellcome Global Monitor⁷, we find that trust is weakly correlated with the Gini inequality index (i.e., trust is higher in countries with more income inequality). One possible explanation for the discrepancy between the Monitor and our study is that urban populations—which are more likely to trust scientists (Fig. 2)—were overrepresented in our samples from countries with high Gini scores, e.g., South Africa. However, a non-preregistered analysis advised against this explanation: The extent of oversampling urban participants (difference of urban-residence individuals in sample vs. in population) did not moderate the effect of the Gini index on trust in scientists. We found tentative support for another explanation: The relationship between income inequality and trust (fig. S13) is largely driven by countries with a high degree of corruption (primarily Latin American countries as well as Sub-Saharan African countries), as indicated by a significant, but very low-powered ($1 - \beta = 0.25$ at $\alpha = .05$) interaction effect of the Gini index $\times$ Transparency International’s Corruption Perceptions Index⁶¹ (table S3). This suggests that people in countries with high inequality may see scientists as a trustworthy alternative to perceivably corrupt governments and political and economic elites^{62–64}. Comparing trust in scientists to trust in the national government (based on country estimates from the Wellcome Global Monitor) supports this assumption. Some countries with higher perceived corruption rank considerably lower in

trust in the government than in trust in scientists, whereas the opposite applies to less corrupt countries with lower perceived corruption (see fig. S14). Overall, trust in the government and trust in scientists are not significantly correlated at the country level ($r = 0.138$, $t = 1.104$, $df = 63$, $p = .274$). We do not find credible evidence that trust is higher in countries with higher average science literacy scores and government expenditures on education, which challenges assumptions that public understanding of science, and policy measures to increase such understanding, foster trust in scientists⁶⁵.

Normative perceptions of scientists' role in society and policymaking

Left-right divides in public opinion about science often centre on the question of whether scientists should take an active role in policymaking⁶⁶. We find that people tend to agree that scientists should engage in society and policymaking (global $M = 3.64$, $SD = 0.87$, 1 = strongly disagree – 5 = strongly agree). Globally, a large majority (83%) agrees that scientists should communicate about science with the public, particularly in African countries. Globally, only a minority disagrees that scientists should actively advocate for specific policies (23%), communicate their findings to politicians (19%) and be more involved in the policymaking process (21%). However, perceptions differ across countries (fig. S15).

About a quarter of the global sample selected the scale midpoints, therefore neither agreeing nor disagreeing on whether scientists should be more involved in policymaking and society (Fig. 4). People with high trust in scientists strongly favour scientists' engagement in society and policymaking ($b = 0.262$, $p < .001$), especially in English-speaking countries (fig. S16). Support for scientists' engagement in society and policymaking also varies both between and within countries. People who are younger, have tertiary education and higher income, or live in urban areas, generally approve of scientists' engagement in policymaking (table S4). Also, right-leaning people and conservatives disapprove of scientists' engagement in society and policymaking.

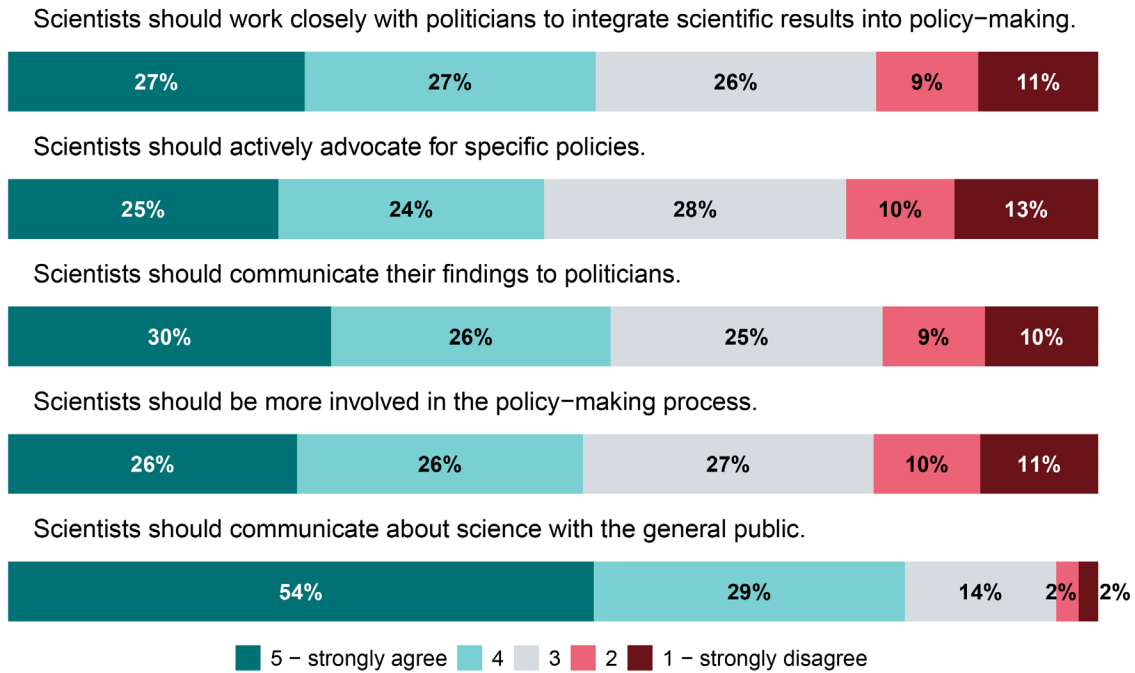


Fig. 4. Normative perceptions of scientists in society and policymaking using weighted response probabilities.

Desired research priorities and perceptions that these are tackled

We hypothesised that trust in scientists relates to another normative belief about science: Expectations about which societal goals scientists should prioritise⁶⁷. We compared whether people's expectations match their perceptions of whether scientists actually tackle the following goals: improving public health, solving energy problems, reducing poverty, and developing defence and military technology.

Globally, people assign the highest priority to improving public health ($M = 4.49$, $SD = 0.84$, 1 = low – 5 = high), followed by solving energy problems ($M = 4.38$, $SD = 0.90$), and reducing poverty ($M = 4.08$, $SD = 1.10$). Responses suggest a substantial discrepancy between what they perceive science is currently prioritising and what they expect scientists to prioritise, with poverty reduction showing the most substantial discrepancy (Fig. 5). The least desired research goal is developing defence and military technology ($M = 3.10$, $SD = 1.36$). Again, there are large differences between global regions (ranging from $M = 1.88$, $SD = 1.21$ in Uruguay to $M = 4.07$, $SD = 1.52$ in the Democratic Republic of Congo). In African and Asian countries, people often demand high priority for developing defence and military technology, as opposed to people in most European and Latin American countries (fig. S17). Overall, people tend to think science prioritises developing defence and military technology more than they desire.

Perceived and desired priorities of scientific research

How strongly do you believe that scientists should prioritize these goals and that science actually tackles them?

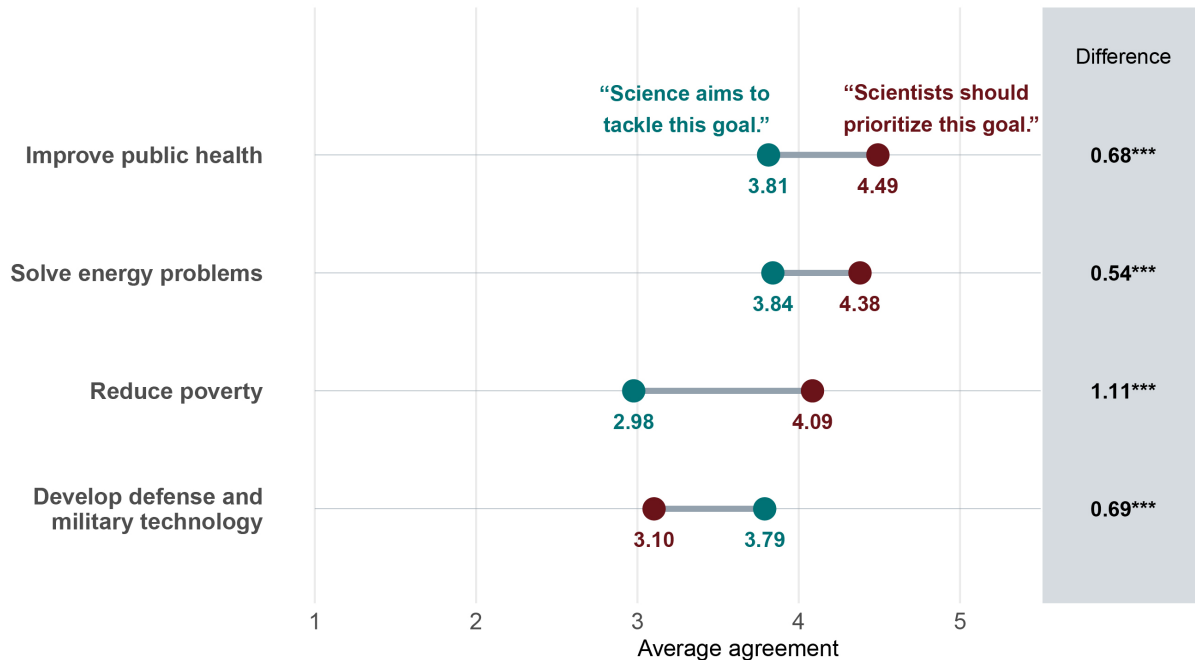


Fig. 5. Perceived research priorities for four goals of scientific research (blue) and desired research priorities (red). Grey horizontal lines indicate the discrepancy between perceived research priorities (“Science aims to tackle this goal”) and desired priorities (“Scientists should prioritize this goal”). Stars indicate results of weighted paired-samples t-tests for significant differences between perceptions and expectations, *** $p < .001$.

Further analyses show that the discrepancy between people’s perceived and desired research priorities is associated with trust in scientists. On the one hand, higher trust is linked to a lower likelihood that scientists’ efforts do not meet people’s expectations for the following goals: improve public health, solve energy problems, and reduce poverty (table S5; see exploratory analyses with reversed hypothesised causality in table S6). In other words, the more people trust scientists, the more they perceive that scientists’ efforts exceed expectations. On the other hand, a higher likelihood that peoples’ perceived efforts exceed expectations is associated with less trust in scientists in the case of developing defence and military technology (i.e., those who think scientists are too focused on defence and military technology trust science less).

Discussion and recommendations

Our global, 68-country survey challenges the idea that there is a widespread lack of public trust in scientists. In most countries, scientists and scientific methods are trusted. This finding is in

line with other international studies on trust in scientists^{6–8}. Our study thus confirms, expands and strengthens previous work that refutes the narrative of a wide-ranging crisis of trust. We expand previous studies by providing the largest dataset on trust in scientists post-pandemic and relying on a theoretically informed multidimensional trust measure. We also show that certain factors, such as being male, conservative, having a social dominance orientation and science-populist attitudes, are correlated with lower trust in scientists.

Public perception of scientific integrity—one of four components of trust—is high, but perceptions of scientists’ openness are comparably lower. Therefore, scientists wishing to gain more public trust could work on being more receptive to feedback, more transparent about their funding and data sources, and invest more effort into communicating about science with the public—which we found to be desired by 83% of respondents. We recommend avoiding top-down communication but encouraging public participation in genuine dialogue, in which scientists seek to consider the insights and needs of other societal actor⁶⁸. While our cross-sectional data cannot speak to the causality of these recommendations, they are supported by the extensive literature on the determinants of trust in scientists (see e.g., ^{13,25,69}).

Trust differs considerably across countries (Fig. 1, fig. S18-21), and there is substantial variation of the trust dimensions, which demonstrates the importance of using multidimensional trust measures like ours in comparative survey research²⁵. This, in turn, will help scientists and science communicators to better understand how to act in ways that increase different components of trustworthiness perceptions, i.e., competence, integrity, benevolence, and openness. Relatedly, trust in scientists varies across population groups, with women, older and more educated people trusting scientists more. While demographic characteristics likely cannot cause views about scientists per se, they reflect shared direct or mediated experiences with scientists. For example, women’s lived experiences with science are likely different from those of men. Media coverage disproportionally features male scientists and portrays them differently than female scientists, which may evoke different trustworthiness perceptions across genders^{70,71}.

This information can help scientists and science communicators to better tailor their communication to different audiences. Our study assessed trust in scientists without distinguishing between different scientific fields. In some countries, trust may depend on the scientists’ discipline and the potential impacts of science on public policy^{18,72}.

Trust and its correlates such as right-leaning and conservative political orientation, education and religiosity clearly vary across countries. This exemplifies the need for more international research that includes underrepresented countries and understudied subpopulations. Our findings also highlight the inconsistency in the association between political orientation depending on the measure used (left-right versus liberal-conservative) and country of study, as well as the importance of ideology—specifically SDO—in relating to trust in scientists. While previous research finds that SDO is associated with the rejection of specific scientific information, such as the reality of climate change⁷³ or the safety and efficacy of vaccines²⁶, our

findings support the idea that SDO may play a more fundamental role in undermining trust in scientists more generally. Scientists who challenge unjust social hierarchies might increase benevolence perceptions among some groups but would likely further decrease trust among people with social dominance orientations. Interventions could be developed to build perceived trustworthiness of scientists and involve trusted communicators outside of scientific institutions.

While no country has low trust in scientists on average, lack of trust in scientists by even a small minority needs to be taken seriously. Distrusting minorities may affect considerations of scientific evidence in policymaking, as well as decisions by individuals that can affect society at large, especially if they receive extensive news media coverage and include people in positions of power that can influence policymaking^{11,74,75}. A minority of 10% can be sufficient to flip a majority⁷⁶ and when a critical mass value of 25% is reached, majority opinion could be tipped⁷⁷. In the context of climate change, an agent-based model showed that an evidence-resistant minority can delay the process of public opinion converging with the scientific consensus⁷⁸. Future research should investigate the size of these distrusting minorities across countries and their characteristics.

In most countries, a majority of people want scientists to take part in policymaking. Future global comparative research should analyse whether opinions differ depending on a scientist's expertise regarding a policy issue⁷⁹ and public support for the policy in question^{15,80,81}. Future studies should also examine whether normative perceptions of science in policymaking shift when specific scientific disciplines or policy issues are mentioned in real-world settings.

A majority of the public want scientists to prioritise research on public health and solving energy problems. Yet, most people believe that scientists are currently not tackling these issues sufficiently and think that defence and military technology are prioritised too much. As the perceived benefits of science are strongly correlated with trust in scientists, greater consideration of public research priorities by scientists, funding agencies and philanthropists, presents an important avenue to increase trust. At the same time, science communication efforts could increasingly focus on highlighting ongoing research on public health and solving energy problems to elevate the prominence of this research in the minds of the public. Our study has several limitations, mostly related to the global data collection effort. For example, our survey was fielded in English/French in some countries where English/French are not the most commonly spoken languages, including Botswana, Ethiopia, Ghana, India, Kenya, and Nigeria. This likely resulted in the oversampling of more educated population segments. This limitation arises from the fact that our survey was fielded as an online survey, which considerably limited the representability of populations in certain countries with lower internet penetration. We provide a more detailed discussion of our limitations in the SI.

Newspapers, opinion pieces, and books³ have spread narratives of low public trust in scientists. However, such claims remain largely unsubstantiated by empirical evidence⁶⁻⁸. Our Many Labs study provides decision-makers, scientists, and the public with large-scale and open

public opinion data on trust in scientists that can help these stakeholders maintain, and potentially increase, trust in scientists.

Online Methods

Overview

The data underlying the analyses were collected in a global, pre-tested, pre-registered, cross-sectional online survey ($N = 71,922$ participants in $k = 68$ countries) between November 2022 and August 2023 as part of the TISP Many Labs project (“Trust in Science and Science-Related Populism”). TISP is an international, multidisciplinary consortium of 241 researchers from 169 institutions across all continents. Researchers conducted surveys within 88 post-hoc weighted quota samples in 68 countries, using the same questionnaire translated into 37 languages. In the following, we will describe the procedures used to collect and analyse the data. Further details are available in the SI and Mede et al.⁴⁶.

Institutional Review Board (IRB) Approval

Our questionnaire was considered exempt from full IRB review from the Harvard University-Area Committee on the Use of Human Subjects (protocol # IRB22-1046) in August 2022. A modified IRB application was submitted and considered exempt from full IRB review by the Harvard University-Area Committee on the Use of Human Subjects in November 2022 (protocol # IRB22-1046). All authors have informed themselves whether IRB approval was required from their institutions and obtained IRB approval if necessary.

Pre-test

A pre-test with $n = 401$ was conducted in the United States in October 2022 to validate the measures used in the questionnaire. Average completion time was 14 minutes. The questionnaire was slightly modified and two questions were added to the survey after the pre-test. Data from the pre-test was not included in the final analyses.

Questionnaire

In total, we measured 111 variables. No identifiable information was collected. The complete questionnaire (in English) is available at the Open Science Framework: [URL REDACTED] The core questionnaire contained the components described in the following. Participants were presented with these components in the order in which they are explained below, but the order of questions and items of multi-item scales was randomised.

Consent form

Participants were asked to carefully read a consent form (approved under IRB protocol # IRB22-1046), which included some general information about the study and the anonymity of the data.

Demographic data - Part I

Participants who consented to participating in the study were then asked to indicate their gender (0 = Female, 1 = Male, 2 = Prefer to self-describe, 99 = prefer not to say), age, and level of education (1 = Did not attend school, 2 = Primary education, 3 = Secondary education (e.g., high school), 4 = Higher education (e.g., university degree or higher education diploma)).

Attention check I

The first attention check asked participants to write the number “213” into a comment box. Participants who failed the attention check were redirected to the end of the survey and were not remunerated. See SI for details on how many respondents failed this attention check in the overall sample and across countries.

Definition of science and scientists

Participants were presented with a definition of science and scientists: *When we say “science”, we mean the understanding we have about the world from observation and testing. When we say “scientists”, we mean people who study nature, medicine, physics, economics, history, and psychology, among other things.* This definition was based on the Wellcome Global Monitor⁷. We added it because in-depth interviews conducted by the Monitor⁷ suggested that including a definition would improve the reliability of cross-country comparisons.

Media use and engagement with science

Participants were asked how often they had come across information about science in ten different places (e.g., in news articles in printed newspapers or magazines) and how often they had engaged with science in four different ways (e.g., have conversations with friends) in the last 12 months (1 = *never*, 7 = *once or more per day*).

Open-ended questions

Participants were randomly assigned to answer one of two open-ended questions. One question asked participants to write who they think benefits the most from science and why, and the second question asked about their opinion on what makes a scientist trustworthy.

Perceived benefits of science

Participants were asked how much they perceived that scientific research benefits people like themselves in their country (1 = not at all, 5 = very strongly), and which geographic region benefits the most and the least from the work that scientists do (1 = Africa, 2 = Asia, 3 = Australia and Oceania, 4 = Europe, 5 = Latin America, 6 = North America).

Desired and perceived goals of science

Participants were asked what goals scientists should prioritize (four items, 1 = very low priority, 5 = very high priority), and how strongly they believed that science aims to tackle these goals (1 = not at all, 5 = very strongly).

Normative perceptions of science and society

Participants rated their agreement to six statements (e.g., scientists should be more involved in the policy-making process) (1 = strongly disagree, 5 = strongly agree). Five of these statements were taken from⁶⁶.

Willingness to be vulnerable to scientists

Participants' willingness to be vulnerable to scientific guidance was assessed with three items (1 = not at all, 5 = very strongly). Willingness to be vulnerable has been conceptualized as a behavioural trust measure as it reflects the ceding of authority⁴⁸.

Trust in scientists

Trust in scientists was assessed with 12 questions on four different dimensions of trustworthiness (i.e., competence, integrity, benevolence, openness) (1 = very [unqualified], 5 = very [qualified]), based on Besley et al.⁴⁸. Psychometric analyses (e.g., scale reliability, exploratory and confirmatory factor analyses, measurement invariance tests) can be found in the SI.

Trust in scientific methods

Participants indicated their level of agreement on whether scientific research methods are the best way to find out if something is true or false (1 = strongly disagree, 5 = strongly agree).

General trust in scientists

A single question taken from Funk et al.⁸² was used to measure participants' level of confidence in scientists (1 = no confidence at all, 5 = a great deal of confidence).

Outspokenness about science

Three items based on McKeever et al.⁸³ were used to measure outspokenness about science (e.g., I will share my opinions about scientific issues, regardless of what others think of them) (1 = strongly disagree, 5 = strongly agree).

Science-related populism

Science-related populist attitudes were assessed with the SciPop Scale³⁸, which comprises eight items (1 = strongly disagree, 5 = strongly agree).

Attention check II

In the second attention check, participants were instructed to select “strongly disagree” to a question. Participants who did not select “strongly disagree” were redirected to the end of the survey and were not remunerated. See SI for details on how many respondents failed this attention check in the overall sample and across countries.

Social dominance orientation

To assess social dominance orientation, we used four items from Pratto et al.⁸⁴ (1 = extremely opposed, 10 = extremely favour).

Trust in climate scientists

Participants were asked how much they trust in scientists in their country who work on climate change (1 = not at all, 5 = very strongly).

Emotions about climate change

Nine different emotions (e.g., helpless) were assessed (1 = not at all, 5 = very strongly).

Perception of government action on climate change

Participants were asked about their perception of government action on climate change with seven items (1 = strongly disagree, 5 = strongly agree)⁸⁵.

Support for climate policies

Participants indicated their support for five policies (1 = not at all, 3 = very much; 4 = not applicable, which was recoded as missing).

Perceptions of extreme weather events

Participants indicated to what extent they believe that climate change has increased the impact of six weather events over the last decades (1 = not at all, 5 = very much). They also indicated whether they expected that climate change will increase the impact of these events in the future (1 = not at all, 5 = very much).

Demographic data - Part II

Participants indicated their household’s yearly net income (in local currency), their political orientation on a spectrum from liberal to conservative (1 = strongly liberal, 5 = strongly conservative, 99 = I don’t know) and on a spectrum from left-leaning to right-leaning (1 = strongly left-leaning, 5 = strongly right-leaning, 99 = I don’t know), their religiosity (1 = not religious at all, 5 = very strongly religious), and whether they live in a rural or urban area. Collaborators were allowed to add questions at the end of the survey. Additional questions did not have to be approved by the lead author.

Translations

The original English survey was translated into the local language where necessary. Translations were done by native speakers who were familiar with the study background and, in many cases, had expertise on survey research and/or the conceptual underpinning of the measures. Minor linguistic adjustments were made to the survey if deemed necessary. Major changes in the wording of the original survey instrument had to be approved by the project lead. In total, the survey instrument was translated into 36 languages and dialects⁴⁶.

Preregistration

We submitted a comprehensive preregistration prior to the data collection to the Open Science Framework (OSF) on 15 November 2022. It included detailed descriptions of our research questions and hypotheses, instruments, data collection, and analytical procedures: [URL REDACTED] Please see SI for deviations from the preregistration.

Power analysis

To determine our minimum target sample size, we ran simulation-based power analyses using the R package *simr* (v1.0.7)⁸⁶, which is designed to conduct power analyses for generalized linear mixed models such as those we used in the main study (for detailed information see SI).

Data collection

Procedure and final sample

Data were collected in surveys that used quotas for age (five bins: 20% 18-29 years, 20% 30-39 years, 20% 40-49 years, 20% 50-59 years, 20% 60 years and older) and gender (two bins: 50% male, 50% female). Participants had to be 18 years of age or older and provide informed consent to participate in the study. Data were collected between November 2022 and August 2023. See Mede et al.,⁴⁶ for an overview of survey periods across countries. The median completion time was 18 minutes.

The surveys were programmed in Qualtrics. Participants that completed the survey were remunerated according to the market research company's local rates. All data was collected via online surveys, except for the Democratic Republic of Congo, where participants were interviewed in face-to-face interviews and responses recorded in Qualtrics by the interviewers. Collaborators were instructed to work with the market research company Bilendi & ResponDi, except for most African countries, where collaborators collected data with MSi. Convenience samples were not accepted.

A total of $n = 72,135$ individuals from 88 samples across $k = 68$ countries completed the survey ($n = 71,922$ after exclusion of duplicate respondents). See Mede et al.⁴⁶ for an overview of all included countries and valid sample sizes across countries (i.e., after exclusion of duplicate respondents) and SI for detailed characteristics of the final sample and the representativeness of surveyed countries by income and regions (Tables S12-13).

Preparing the dataset

Exclusion of non-completes and data quality test

We excluded all respondents who did not complete the survey, because they cancelled participation during the survey, were filtered as their gender $\times$ age quota was already full, or because they did not pass one of the two attention checks. 4.24% of respondents who reached the first attention check did not pass it (“Please write the number 213 into the comment box”). 24.38% of respondents who reached the second attention check did not pass it (“To show us that you are still paying attention, please select ‘strongly disagree’”; ⁴⁶). We excluded all respondents who managed to complete the survey more than once despite countermeasures (e.g., IP address checks). In total, we excluded 213 duplicate respondents.

Outlier value removal

We removed extreme outlier values for age and household income: Age outliers were defined as values smaller than 18 and bigger than 100. Income outliers were defined as values that were smaller than zero, equal to zero, or outside $5 \times$ the interquartile range of the log-transformed income distribution within each country after exclusion of values smaller than zero or equal to zero. This led to the removal of the age values of 8 respondents and the removal of the income values of 2,457 respondents (1,365 respondents indicated income values equal to or smaller than 0; 1,092 respondents indicated income values outside $5 \times$ the interquartile range of the log-transformed income distribution within each country after exclusion of values equal to or smaller than 0).

Post-hoc weighting

We computed post-stratification weights with the R package *survey* (v4.4-2)⁸⁷ to ensure that our models would estimate parameters that are representative for target populations in terms of gender, age, and education and have more precise standard errors (SEs). We used raking⁸⁸ to compute four kinds of weights, i.e. (A) post-stratification weights at country level, (B) sample size weights for each country, (C) post-stratification weights for the complete sample, and (D) rescaled post-stratification weights for multilevel analyses⁴⁶.

Scale reliability

Scales were combined into indices and psychometric properties were assessed for all indices (see SI), including scale reliability (Cronbach’s Alpha and Omega) and cross-country measurement invariances. Scale reliability was good for all scales⁴⁶.

Analyses

Factors explaining trust in scientists

To investigate explanatory factors of trust in scientists and explore how their influence varies across countries, we ran a block-wise linear multilevel regression analysis with the R package lme4 (v1.1-35)⁸⁹. The model included rescaled post-stratification weights⁹⁰.

All independent variables in the first, second, and third block were scaled by country means and country SDs. All independent variables in the fourth block were scaled by grand means and grand SDs.

We first tried to fit a model with random intercepts and random effects for all independent variables. However, this model failed to converge with three negative eigenvalues and also had a singular fit, i.e., some random effects correlations were close to -1/+1, and some random effects variances were close to 0. This was likely because the random effects structure was too complex. Therefore, we simplified the model as follows: To test global effects of the independent variables on trust in scientists, we fitted a model that contained random intercepts across countries (but no random effects) and inspected fixed effects estimates. To investigate how the influence of independent variables varies across countries, we fitted separate models, each of which contained random intercepts across countries and random effects for one particular independent variable. For example, we fitted a regression model with random intercepts across countries and random effects for political orientation (but no random effects for all other independent variables) to assess how the effect of political orientation on trust in scientists varies across countries. This entire procedure was completely in line with our preregistration. Before we fitted the multilevel models, we confirmed that they would fit the data better than fixed-effects models. First, we inspected intra-class correlations for trust in scientists (ICC = 0.170). Second, we ran a likelihood-ratio tests: It showed that a random-intercept null model explaining trust in scientists had significantly better fit than a fixed-effects null model ($\chi^2 = 6,024.9, p < .001$).

Moreover, we tested for multicollinearity of independent variables for the most complex model, i.e., after inclusion of all three blocks of independent variables (see table S16). All variance inflation factors (VIF) were well below even a very conservative threshold value of 4⁹¹.

Normative perceptions of science in policymaking

To examine whether the public demands that scientists should take an active role in society and policymaking, we ran two analyses: First, we computed weighted probabilities of responses to the five items measuring these perceptions. This analysis provided estimates that are approximately representative with regards to gender, age, education, and country sample size. Second, we tested explanatory factors of normative perceptions of science in policy-making and society: We fitted a linear multilevel regression model with the R package lme4⁸⁹, which explained the average agreement to the five individual items measuring those perceptions, included the rescaled post-stratification weights, and contained trust in scientists, science-related

populist attitudes, and sociodemographic variables as independent variables, i.e. gender (binary; 1 = male), age (continuous), education (binary; 1 = tertiary education), annual household income in US dollar (continuous, log-transformed), place of residence (binary; 1 = urban), right-leaning political orientation (continuous), conservative political orientation (continuous), and religiosity. All independent variables were scaled by country means and country SDs.

We specified random intercepts across countries and random effects for trust in scientists and science-related populist attitudes. Significance tests of regression estimates relied on the Satterthwaite method⁹². Before we fitted the multilevel model, we confirmed that it would fit the data better than a fixed-effects model. First, we inspected the intra-class correlation of the normative perceptions index (ICC = 0.104). Second, we ran a likelihood-ratio test, which showed that a random-intercept null model had significantly better fit than a fixed-effects null model ($\chi^2 = 3780.6, p < .001$). Moreover, we tested for multicollinearity of independent variables (see table S17). All VIF values were well below even a very conservative threshold value of 4⁹¹.

Perceived and desired priorities of scientific research

To explore desires that scientists should prioritize four specific goals (improving public health, solving energy problems, reducing poverty, developing defences and military technology) as well as perceptions that science actually tackles these goals, we ran three analyses: First, we inspected weighted mean values of responses to the four items measuring priority desires as well as weighted mean values of responses to the four items measuring perceptions that science actually devote efforts to the four goals.

Second, we ran weighted paired-samples *t*-tests to analyse if mean values of desires and perceptions differed significantly from each other. These analyses provided estimates that are approximately representative with regards to gender, age, education, and country sample size. Third, we tested explanatory factors of the discrepancy between the desire that scientists should prioritize the four goals and perceptions that science actually tackles them. To do so, we ran four block-wise linear multilevel regression analyses with the R package lme4⁸⁹. Each model explained the discrepancy between desires that scientists should prioritize one of the four goals and perceptions that science actually tackles them, with higher outcome variable values indicating that perceptions are more likely to exceed desires and lower outcome variable values indicating that perceptions are more likely to stay behind desires. All models included rescaled post-stratification weights⁹⁰.

For each of the four models, we specified random intercepts across countries and random effects for trust in scientists and science-related populist attitudes. Significance tests of regression estimates relied on the Satterthwaite method⁹². Before we fitted the multilevel models, we confirmed that they would fit the data better than fixed-effects models. First, we inspected the intra-class correlations of the four discrepancy scores (health: ICC = 0.112; energy: ICC = 0.079; poverty: ICC = 0.134; defence: ICC = 0.107). Second, we ran likelihood-ratio tests, which showed that random-intercept null models had significantly better fit than fixed-effects null models (health: $\chi^2 = 3,246.2, p < .001$; energy: $\chi^2 = 2,264.4, p < .001$; poverty: $\chi^2 = 4,835.2,$

$p < .001$; defence: $\chi^2 = 3,669.0$, $p < .001$). Moreover, we tested for multicollinearity of independent variables for the most complex models (see table S18). All VIF values were well below even a very conservative threshold value of 4⁹¹.

References:

1. Sturgis, P., Brunton-Smith, I. & Jackson, J. Trust in science, social consensus and vaccine confidence. *Nat Hum Behav* **5**, 1528–1534 (2021).
2. Algan, Y., Cohen, D., Davoine, E., Foucault, M. & Stantcheva, S. Trust in scientists in times of pandemic: Panel evidence from 12 countries. *Proceedings of the National Academy of Sciences* **118**, e2108576118 (2021).
3. Nichols, T. M. *The Death of Expertise: The Campaign against Established Knowledge and Why It Matters*. (Oxford University Press, New York, NY, 2017).
4. Cologna, V. & Siegrist, M. The role of trust for climate change mitigation and adaptation behaviour: A meta-analysis. *Journal of Environmental Psychology* **69**, 101428 (2020).
5. Cologna, V., Hoogendoorn, G. & Brick, C. To strike or not to strike? an investigation of the determinants of strike participation at the Fridays for Future climate strikes in Switzerland. *PLOS ONE* **16**, e0257296 (2021).
6. IPSOS. Ipsos Global Trustworthiness Monitor: Stability in an Unstable World. (2022).
7. Wellcome Global Monitor. Wellcome Global Monitor: How does the world feel about science and health? <https://wellcome.org/sites/default/files/wellcome-global-monitor-2018.pdf> (2018).
8. Wellcome Global Monitor. Wellcome Global Monitor: How Covid-19 affected people's lives and their views about science. <https://cms.wellcome.org/sites/default/files/2021-11/Wellcome-Global-Monitor-Covid.pdf> (2020).
9. Jamieson, K. H. Crisis or self-correction: Rethinking media narratives about the well-being of science. *Proceedings of the National Academy of Sciences* **115**, 2620–2627 (2018).
10. Shanahan, E. A., Mcbeth, M. K. & Hathaway, P. L. Narrative Policy Framework: The Influence of Media Policy Narratives on Public Opinion. *Politics & Policy* **39**, 373–400 (2011).
11. Toff, B. Exploring the effects of polls on public opinion: How and when media reports of policy preferences can become self-fulfilling prophecies. *Research & Politics* **5**, 2053168018812215 (2018).
12. Sonck, N. & Loosveldt, G. Impact of Poll Results on Personal Opinions and Perceptions of Collective Opinion. *International Journal of Public Opinion Research* **22**, 230–255 (2010).
13. Hendriks, F., Kienhues, D. & Bromme, R. Measuring Laypeople's Trust in Experts in a Digital Age: The Muenster Epistemic Trustworthiness Inventory (METI). *PLOS ONE* **10**, e0139309 (2015).
14. Wintterlin, F. *et al.* Predicting Public Trust in Science: The Role of Basic Orientations Toward Science, Perceived Trustworthiness of Scientists, and Experiences With Science. *Frontiers in Communication* **6**, (2022).
15. Cologna, V., Baumberger, C., Knutti, R., Oreskes, N. & Berthold, A. The Communication of Value Judgements and its Effects on Climate Scientists' Perceived Trustworthiness. *Environmental Communication* **16**, 1094–1107 (2022).

16. Bromme, R., Mede, N. G., Thomm, E., Kremer, B. & Ziegler, R. An anchor in troubled times: Trust in science before and within the COVID-19 pandemic. *PLOS ONE* **17**, e0262823 (2022).
17. Motta, M. The enduring effect of scientific interest on trust in climate scientists in the United States. *Nature Clim Change* **8**, 485–488 (2018).
18. McCright, A. M., Dentzman, K., Charters, M. & Dietz, T. The influence of political ideology on trust in science. *Environ. Res. Lett.* **8**, 044029 (2013).
19. Azevedo, F. & Jost, J. T. The ideological basis of antiscientific attitudes: Effects of authoritarianism, conservatism, religiosity, social dominance, and system justification. *Group Processes & Intergroup Relations* **24**, 518–549 (2021).
20. Mede, N. G. Legacy media as inhibitors and drivers of public reservations against science: global survey evidence on the link between media use and anti-science attitudes. *Humanit Soc Sci Commun* **9**, 1–11 (2022).
21. Rutjens, B. T. *et al.* Science Skepticism Across 24 Countries. *Social Psychological and Personality Science* **13**, 102–117 (2022).
22. Chan, E. Are the religious suspicious of science? Investigating religiosity, religious context, and orientations towards science. *Public Underst Sci* **27**, 967–984 (2018).
23. Gil de Zúñiga, H., Ardèvol-Abreu, A., Diehl, T., Gómez Patiño, M. & Liu, J. Trust in Institutional Actors across 22 Countries. Examining Political, Science, and Media Trust Around the World. (2019) doi:10.4185/RLCS-2019-1329en.
24. Siegrist, M., Cvetkovich, G. & Roth, C. Salient Value Similarity, Social Trust, and Risk/Benefit Perception. *Risk Analysis* **20**, 353–362 (2000).
25. Besley, J. C. & Tiffany, L. A. What are you assessing when you measure “trust” in scientists with a direct measure? *Public Underst Sci* 09636625231161302 (2023) doi:10.1177/09636625231161302.
26. Ghai, S., Forscher, P. S. & Chuan-Peng, H. The illusion of generalizability in one big team science study. Preprint at <https://doi.org/10.31234/osf.io/avcsp> (2023).
27. Forscher, P. S. *et al.* The Benefits, Barriers, and Risks of Big-Team Science. *Perspect Psychol Sci* **18**, 607–623 (2023).
28. Odeny, B. & Bosurgi, R. Time to end parachute science. *PLoS Med* **19**, e1004099 (2022).
29. Dubow, S. Racial irredentism, ethnogenesis, and white supremacy in high-apartheid South Africa. *Kronos* **41**, 236–264 (2015).
30. Brandt, A. M. Racism and Research: The Case of the Tuskegee Syphilis Study. *The Hastings Center Report* **8**, 21–29 (1978).
31. Scharff, D. P. *et al.* More than Tuskegee: Understanding Mistrust about Research Participation. *J Health Care Poor Underserved* **21**, 879–897 (2010).
32. West, J. D. & Bergstrom, C. T. Misinformation in and about science. *Proceedings of the National Academy of Sciences* **118**, e1912444117 (2021).
33. Roozenbeek, J. *et al.* Susceptibility to misinformation about COVID-19 around the world. *Royal Society Open Science* **7**, 201199 (2020).
34. Hendriks, F., Kienhues, D. & Bromme, R. Replication crisis = trust crisis? The effect of successful vs failed replications on laypeople’s trust in researchers and research. *Public Underst Sci* **29**, 270–288 (2020).
35. Rutjens, B. T. & Većkalov, B. Conspiracy beliefs and science rejection. *Current Opinion in Psychology* **46**, 101392 (2022).

36. Douglas, K. M. Are Conspiracy Theories Harmless? *The Spanish Journal of Psychology* **24**, e13 (2021).
37. Mede, N. G. & Schäfer, M. S. Science-related populism: Conceptualizing populist demands toward science. *Public Underst Sci* **29**, 473–491 (2020).
38. Mede, N. G., Schäfer, M. S. & Fuchslin, T. The SciPop Scale for Measuring Science-Related Populist Attitudes in Surveys: Development, Test, and Validation. *International Journal of Public Opinion Research* **33**, 273–293 (2021).
39. Funk, C., Tyson, A., Kennedy, B. & Johnson. Science and Scientists Held in High Esteem Across Global Publics. *Pew Research Center Science & Society* <https://www.pewresearch.org/science/2020/09/29/science-and-scientists-held-in-high-esteem-across-global-publics/> (2020).
40. Li, N. & Qian, Y. Polarization of public trust in scientists between 1978 and 2018: Insights from a cross-decade comparison using interpretable machine learning. *Politics and the Life Sciences* **41**, 45–54 (2022).
41. Sidanius, J. & Pratto, F. *Social Dominance: An Intergroup Theory of Social Hierarchy and Oppression*. (Cambridge University Press, Cambridge, 1999). doi:10.1017/CBO9781139175043.
42. Sidanius, J., Pratto, F., Martin, M. & Stallworth, L. M. Consensual Racism and Career Track: Some Implications of Social Dominance Theory. *Political Psychology* **12**, 691–721 (1991).
43. Kerr, J. R. & Wilson, M. S. Right-wing authoritarianism and social dominance orientation predict rejection of science and scientists. *Group Processes & Intergroup Relations* **24**, 550–567 (2021).
44. Achterberg, P., de Koster, W. & van der Waal, J. A science confidence gap: Education, trust in scientific methods, and trust in scientific institutions in the United States, 2014. *Public Underst Sci* **26**, 704–720 (2017).
45. Besley, J. C., Lee, N. M. & Pressgrove, G. Reassessing the Variables Used to Measure Public Perceptions of Scientists. *Science Communication* 1075547020949547 (2020) doi:10.1177/1075547020949547.
46. Mede, N. G. *et al.* Perceptions of Science, Science Communication, and Climate Change Attitudes in 67 Countries: The TISP Dataset. (2024) doi:10.31234/osf.io/jktsy.
47. Putnick, D. L. & Bornstein, M. H. Measurement invariance conventions and reporting: The state of the art and future directions for psychological research. *Developmental Review* **41**, 71–90 (2016).
48. Besley, J. C., Lee, N. M. & Pressgrove, G. Reassessing the Variables Used to Measure Public Perceptions of Scientists. *Science Communication* **43**, 3–32 (2021).
49. Rutjens, B. T. & van der Lee, R. Spiritual skepticism? Heterogeneous science skepticism in the Netherlands. *Public Underst Sci* **29**, 335–352 (2020).
50. McPhetres, J., Jong, J. & Zuckerman, M. Religious Americans Have Less Positive Attitudes Toward Science, but This Does Not Extend to Other Cultures. *Social Psychological and Personality Science* **12**, 528–536 (2021).
51. De Cruz, H. Religion and Science. in *The Stanford Encyclopedia of Philosophy* (eds. Zalta, E. N. & Nodelman, U.) (Metaphysics Research Lab, Stanford University, 2022).
52. Johnson, C., Thigpen, C. & Funk, C. On the Intersection of Science and Religion. *Pew Research Center's Religion & Public Life Project*

- <https://www.pewresearch.org/religion/2020/08/26/on-the-intersection-of-science-and-religion/> (2020).
53. Vlasceanu, M. *et al.* Addressing climate change with behavioral science: A global intervention tournament in 63 countries. *Science Advances* **10**, eadj5778 (2024).
 54. Van Bavel, J. J. *et al.* National identity predicts public health support during a global pandemic. *Nat Commun* **13**, 517 (2022).
 55. De keersmaecker, J., Schmid, K., Sibley, C. G. & Osborne, D. The association between political orientation and political knowledge in 45 nations. *Sci Rep* **14**, 2590 (2024).
 56. Lasco, G. & Curato, N. Medical populism. *Social Science & Medicine* **221**, 1–8 (2019).
 57. Otjes, S. & Rekker, R. Socialised to think in terms of left and right? The acceptability of the left and the right among European voters. *Electoral Studies* **72**, 102365 (2021).
 58. Weeden, J. & Kurzban, R. Do People Naturally Cluster into Liberals and Conservatives? *Evolutionary Psychological Science* **2**, 47–57 (2016).
 59. Thorisdottir, H., Jost, J. T., Liviatan, I. & Shrout, P. E. Psychological Needs and Values Underlying Left-Right Political Orientation: Cross-National Evidence from Eastern and Western Europe. *Public Opinion Quarterly* **71**, 175–203 (2007).
 60. Bauer, P. C., Barberá, P., Ackermann, K. & Venetz, A. Is the Left-Right Scale a Valid Measure of Ideology? *Polit Behav* **39**, 553–583 (2017).
 61. Transparency International. 2022 Corruption Perceptions Index. *Transparency.org* <https://www.transparency.org/en/cpi/2022> (2022).
 62. *The Cultural Authority of Science: Comparing across Europe, Asia, Africa and the Americas*. (Routledge, London, 2018).
 63. Clausen, B., Kraay, A. & Nyiri, Z. Corruption and Confidence in Public Institutions: Evidence from a Global Survey. *The World Bank Economic Review* **25**, 212–249 (2011).
 64. Jong-sung, Y. & Khagram, S. A Comparative Study of Inequality and Corruption. *Am Sociol Rev* **70**, 136–157 (2005).
 65. Sturgis, P. & Allum, N. Science in Society: Re-Evaluating the Deficit Model of Public Attitudes. *Public Underst Sci* **13**, 55–74 (2004).
 66. Cologna, V., Knutti, R., Oreskes, N. & Siegrist, M. Majority of German citizens, US citizens and climate scientists support policy advocacy by climate researchers and expect greater political engagement. *Environ. Res. Lett.* **16**, 024011 (2021).
 67. Besley, J. C. The National Science Foundation’s science and technology survey and support for science funding, 2006–2014. *Public Underst Sci* **27**, 94–109 (2018).
 68. Bubela, T. *et al.* Science communication reconsidered. *Nat Biotechnol* **27**, 514–518 (2009).
 69. Fiske, S. T. & Dupree, C. Gaining trust as well as respect in communicating to motivated audiences about science topics. *PNAS* **111**, 13593–13597 (2014).
 70. Freedman, G., Moutoux, I., Hermans, I. & Green, M. C. “She made a mean beef stroganoff”: Gendered portrayals of women in STEM in newspaper articles and their effects. *Communication Monographs* **91**, 262–282 (2024).
 71. Mitchell, M. & McKinnon, M. ‘Human’ or ‘objective’ faces of science? Gender stereotypes and the representation of scientists in the media. *Public Underst Sci* **28**, 177–190 (2019).
 72. Myers, T. A. *et al.* Predictors of trust in the general science and climate science research of US federal agencies. *Public Understanding of Science* **26**, 843–860 (2017).
 73. Stanley, S. K. & Wilson, M. S. Meta-analysing the association between social dominance orientation, authoritarianism, and attitudes on the environment and climate change. *Journal of Environmental Psychology* **61**, 46–56 (2019).

74. Franta, B. Weaponizing economics: Big Oil, economic consultants, and climate policy delay. *Environmental Politics* **31**, 555–575 (2022).
75. Oreskes, N. & Conway, E. M. *Merchants of Doubt: How a Handful of Scientists Obscured the Truth on Issues From Tobacco Smoke to Global Warming*. (Bloomsbury Press, 2010).
76. Xie, J. *et al.* Social consensus through the influence of committed minorities. *Phys. Rev. E* **84**, 011130 (2011).
77. Centola, D., Becker, J., Brackbill, D. & Baronchelli, A. Experimental evidence for tipping points in social convention. *Science* **360**, 1116–1119 (2018).
78. Lewandowsky, S., Pilditch, T. D., Madsen, J. K., Oreskes, N. & Risbey, J. S. Influence and seepage: An evidence-resistant minority can affect public opinion and scientific belief formation. *Cognition* **188**, 124–139 (2019).
79. Oreskes, N. What Is the Social Responsibility of Climate Scientists? *Daedalus* **149**, 33–45 (2020).
80. Beall, L., Myers, T. A., Kotcher, J., Vraga, E. K. & Maibach, E. W. Controversy matters: Impacts of topic and solution controversy on the perceived credibility of a scientist who advocates. *PLOS ONE* **12**, e0187511 (2017).
81. Kotcher, J., Myers, T. A., Vraga, E. K., Stenhouse, N. & Maibach, E. W. Does Engagement in Advocacy Hurt the Credibility of Scientists? Results from a Randomized National Survey Experiment. *Environmental Communication* **11**, 415–429 (2017).
82. Funk, C., Hefferon, M., Kennedy, B. & Johnson, C. Trust and Mistrust in Americans' Views of Scientific Experts. *Pew Research Center Science & Society* <https://www.pewresearch.org/science/2019/08/02/trust-and-mistrust-in-americans-views-of-scientific-experts/> (2019).
83. McKeever, R., McKeever, B. W. & Li, J.-Y. Speaking up Online: Exploring Hostile Media Perception, Health Behavior, and Other Antecedents of Communication. *Journalism & Mass Communication Quarterly* **94**, 812–832 (2017).
84. Pratto, F. *et al.* Social dominance in context and in individuals: Contextual moderation of robust effects of social dominance orientation in 15 languages and 20 countries. *Social Psychological and Personality Science* **4**, 587–599 (2013).
85. Hickman, C. *et al.* Climate anxiety in children and young people and their beliefs about government responses to climate change: a global survey. *The Lancet Planetary Health* **5**, e863–e873 (2021).
86. Green, P. & MacLeod, C. J. SIMR: an R package for power analysis of generalized linear mixed models by simulation. *Methods in Ecology and Evolution* **7**, 493–498 (2016).
87. Lumley, T. *survey: Analysis of Complex Survey Samples*. (2023).
88. Battaglia, M. P., Hoaglin, D. C. & Frankel, M. R. Practical Considerations in Raking Survey Data. *Survey Practice* **2**, (2009).
89. Bates, D. *et al.* lme4: Linear Mixed-Effects Models using 'Eigen' and S4. (2023).
90. Patil, I. *et al.* datawizard: Easy Data Wrangling and Statistical Transformations. (2023).
91. O'Brien, R. M. A Caution Regarding Rules of Thumb for Variance Inflation Factors. *Qual Quant* **41**, 673–690 (2007).
92. Satterthwaite, F. E. Synthesis of variance. *Psychometrika* **6**, 309–316 (1941).
93. van der Meer, T. & Hakhverdian, A. Political Trust as the Evaluation of Process and Performance: A Cross-National Study of 42 European Countries. *Political Studies* **65**, 81–102 (2017).

94. O'Brien, T. L. & Noy, S. Cultural Authority in Comparative Context: A Multilevel Analysis of Trust in Science and Religion. *Journal for the Scientific Study of Religion* **57**, 495–513 (2018).
95. Vallejo, B. M. & Ong, R. A. C. Policy responses and government science advice for the COVID 19 pandemic in the Philippines: January to April 2020. *Progress in Disaster Science* **7**, 100115 (2020).
96. Tony Blair Institute for Global Change. Pandemic Populism: An Analysis of Populist Leaders' Responses to Covid-19. <https://www.institute.global/insights/public-services/pandemic-populism-analysis-populist-leaders-responses-covid-19> (2020).
97. Bušítková, L. & Baboš, P. Best in Covid: Populists in the Time of Pandemic. *Politics and Governance* **8**, 496–508 (2020).
98. Allen, M. S., Iliescu, D. & Greiff, S. Single Item Measures in Psychological Science. *European Journal of Psychological Assessment* **38**, 1–5 (2022).
99. Tavakol, M. & Dennick, R. Making sense of Cronbach's alpha. *Int J Med Educ* **2**, 53–55 (2011).
100. Streiner, D. L. Starting at the Beginning: An Introduction to Coefficient Alpha and Internal Consistency. *Journal of Personality Assessment* **80**, 99–103 (2003).
101. Wuttke, A., Schimpf, C. & Schoen, H. When the Whole Is Greater than the Sum of Its Parts: On the Conceptualization and Measurement of Populist Attitudes and Other Multidimensional Constructs. *American Political Science Review* **114**, 356–374 (2020).
102. Mede, N. G., Schäfer, M. S., Metag, J. & Klinger, K. Who supports science-related populism? A nationally representative survey on the prevalence and explanatory factors of populist attitudes toward science in Switzerland. *PLOS ONE* **17**, e0271204 (2022).
103. Mede, N. G., Schäfer, M. S. & Metag, J. Cognitio populi – Vox populi: Implications of science-related populism for communication behavior. *Communications* (2023) doi:10.1515/commun-2022-0059.
104. World Bank & OECD. GDP per capita (current US\$). (2022).
105. Directorate-General of Budget, Accounting and Statistics, Republic of China. Statistical Yearbook of the Republic of China 2021: Per capita income and consumption. (2022).
106. UNESCO Institute for Statistics. Government expenditure on education, total (% of GDP). (2022).
107. Directorate-General of Budget, Accounting and Statistics, Republic of China. Statistical Yearbook of the Republic of China 2021: Summary of R&D expenditure. (2022).
108. World Bank. Gini index. Poverty and Inequality Platform. (2022).
109. World Economics. Gini Coefficient. (2022).
110. PISA. Science performance (PISA): Boys / Girls, Mean score, 2018 or latest available. (2022).
111. OECD. Pisa Database. (2022).
112. V-Dem. Country-Year: V-Dem Core Dataset v13. (2023).
113. Spannagel, J. & Kinzelbach, K. The Academic Freedom Index and Its indicators: Introduction to new global time-series V-Dem data. *Qual Quant* (2022) doi:10.1007/s11135-022-01544-0.
114. Norris, P. & Inglehart, R. *Cultural Backlash: Trump, Brexit, and Authoritarian Populism*. (Cambridge University Press, 2019).
115. Devinney, T. M. & Hartwell, C. A. Varieties of populism. *Global Strategy Journal* **10**, 32–66 (2020).

116. Guan, T. & Yang, Y. Rights-oriented or responsibility-oriented? Two subtypes of populism in contemporary China. *International Political Science Review* **42**, 672–689 (2021).
117. Zhang, C. Right-wing populism with Chinese characteristics? Identity, otherness and global imaginaries in debating world politics online. *European Journal of International Relations* **26**, 88–115 (2020).
118. Aslanidis, P. Populism and Social Movements. in *The Oxford Handbook of Populism* (eds. Kaltwasser, C. R., Taggart, P., Espejo, P. O. & Ostiguy, P.) 0 (Oxford University Press, 2017). doi:10.1093/oxfordhb/9780198803560.013.23.
119. Lam-Knott, S. “Populism” in Hong Kong’s contemporary politics. *The Immanent Frame*. <https://tif.ssrc.org/2020/02/26/populism-in-hong-kongs-contemporary-politics/> (2020).
120. Ng, H.-Y. & Kennedy, K. J. Localist groups and populist radical regionalism in Hong Kong. *China: An International Journal* **17**, 111–134 (2019).
121. Tang, G. Media populism in post-handover Hong Kong: an investigation of media framing of public finance. *Chinese Journal of Communication* **10**, 433–449 (2017).
122. Claessens, S., Kyritsis, T. & Atkinson, Q. D. Cross-national analyses require additional controls to account for the non-independence of nations. *Nat Commun* **14**, 5776 (2023).
123. Paccagnella, O. Centering or Not Centering in Multilevel Models? The Role of the Group Mean and the Assessment of Group Effects. *Eval Rev* **30**, 66–85 (2006).
124. Enders, C. K. & Tofighi, D. Centering predictor variables in cross-sectional multilevel models: A new look at an old issue. *Psychological Methods* **12**, 121–138 (2007).
125. Yaremych, H. E., Preacher, K. J. & Hedeker, D. Centering categorical predictors in multilevel models: Best practices and interpretation. *Psychological Methods* **28**, 613–630 (2023).
126. Kenward, M. G. & Roger, J. H. Small Sample Inference for Fixed Effects from Restricted Maximum Likelihood. *Biometrics* **53**, 983–997 (1997).
127. Luke, S. G. Evaluating significance in linear mixed-effects models in R. *Behav Res* **49**, 1494–1502 (2017).

Notes:

ⁱ The term “country” in this article refers to both sovereign states and territories not recognized as such.

ⁱⁱ For example, “science” does not translate precisely into German, Swedish, and Polish, where the term also encompasses the “humanities”.

Acknowledgments: We thank Damiano Lombardi (University of Zurich) for managing the author list and author contributions and Peter Licari (Technite) for valuable methodological advice.

Funding: The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript. Detailed funding information has been removed for this preprint and is available upon request.

Author contributions:

Conceptualization: VC, NM, SB, JB, CB, MJ, EWM, SM, NO, MS, SVDL

Data curation: NM

Formal analysis: NM

Methodology: VC, NM, SB, JB, CB, MJ, EWM, SM, NO, MS, SVDL

Project admin.: VC

Software: VC, NM

Supervision: VC, NM

Validation: VC, NM

Visualization: VC, NM

Investigation: VC, NM, NIAA, SA, NAS, BA, IA, EA, AA, MA, IMA, MA, RMA, RA, TA, DA, FA, AB, RB, KB, EB, AYB, OB, KB, AB, OB, PCA, FC, ACV, TC, RKC, SC, GC, SDP, RD, SD, CDC, LDS, KD, SD, KMD, CD, DD, MD, UKHE, TWE, MF, AFB, ZF, XF, CF, CF, MF, SF, HF, JF, MF, SF, AG, PGV, MG, WG, OG, OG, TG, JG, EG, MG, CGB, HG, DG, GMG, LG, HH, LNH, PH, ACHM, AH, GH, MH, MH, NI, MI, MTI, YJ, TJ, CAJ, SJ, DJ, MZK, JK, SK, JRK, MK, TKR, OK, HK, AK, LK, EK, LK, AK, LSK, PK, SK, TK, AK, EAK, CL, AL, AL, JBL, ZL, NL, AL, GL, AL, ALO, CLV, NML, CHL, KLT, MDM, SM, HM, JM, TLM, JM, PM, FMR, MM, IM, ZM, JN, EJN, JPN, FLNC, NNVN, DN, TO, JPH, MP, PP, PPS, MPC, MP, YP, ARP, MP, KP, MP, JP, DP, AP, KP, EP, KP, DMQ, PR, AR, FGR, CRS, GR, JR, SR, JPR, RMR, IR, OS, RRS, PS, BS, AS, JSN, ES, NS, LS, BS, OS, SKS, GS, SS, BS, ES, MT, CTE, CTE, BT, AKT, RT, DTF, MT, OMU, ICU, JVN, CV, SV, IV, AVB, IW, IW, MW, TW, MW, FW, ADW, ZX, JX, EZP, AZ, RAZ

Resources: NIAA, SA, NAS, BA, IA, EA, AA, MA, IMA, MA, MA, RMA, RA, TA, DA, FA, AB, RB, KB, EB, AYB, OB, KB, AB, OB, PCA, FC, ACV, TC, RKC, SC, GC, SDP, RD, SD, CDC, LDS, KD, SD, KMD, CD, DD, MD, UKHE, TWE, MF, AFB, ZF, XF, CF, CF, MF, SF, HF, JF, MF, SF, AG, PGV, MG, WG, OG, OG, TG, JG, EG, MG, CGB, HG, DG, GMG, LG, HH, LNH, PH, ACHM, AH, GH, MH, MH, NI, MI, MTI, YJ, TJ, CAJ, SJ, DJ, MZK, JK, SK, JRK, MK, TKR, OK, HK, AK, LK, EK, LK, AK, LSK, PK, SK, TK, AK, EAK, CL, AL, AL, JBL, ZL, NL, AL, GL, AL, ALO, CLV, NML, CHL, KLT, MDM, SM, HM, JM, TLM, JM, PM, FMR, MM, IM, ZM, JN, EJN, JPN, FLNC, NNVN, DN, TO, JPH, MP, PP, PPS, MPC, MP, YP, ARP, MP, KP, MP, JP, DP, AP, KP, EP, KP, DMQ, PR, AR, FGR, CRS, GR, JR, SR, JPR, RMR, IR, OS, RRS, PS, BS, AS, JSN, ES, NS, LS, BS, OS, SKS, SS, BS, ES, MT, CTE, CTE, BT, AKT, RT, DTF, MT, OMU, ICU, JVN, CV, SV, IV, AVB, IW, IW, MW, TW, MW, FW, ADW, ZX, JX, EZP, AZ, RAZ

Funding acquisition: VC, NO, MS, JB, EM, SB, CB, BA, IA, EA, MA, MA, RMA, DA, AB, RB, EB, CB, AYB, RB, OB, MB, AB, KB, AB, OB, PCA, FC, ACV, TC, SC, GC, RD, SD, CDC, LDS, KD, SD, KMD, CD, DD, MD, UKHE, ME, BE, TWE, MF, AFB, XF, CF, CF, MF, SF, HF, JF, MF, SF, AG, MG, WG, OG, OG, TG, JG, EG, MG,

CGB, HG, DG, GMG, LG, HH, LNH, PH, ACHM, AH, GH, MH, MH, NI, MI, CAJ, SJ, DJ, MZK, JK, SK, JRK, TKR, OK, HK, TK, AK, LK, EK, LK, LK, AK, JK, LSK, PK, SK, AK, EAK, CL, AL, AL, JBL, ZL, NL, AL, GL, AL, ALO, CLV, NML, CHL, KLT, MDM, SM, RM, HM, JM, TLM, JM, PM, FMR, MM, EJN, JPN, TO, TO, JPH, MP, PP, PPS, MPC, MP, YP, ARP, MP, CRP, KP, JP, DP, AP, EP, KP, PR, AR, FGR, CRS, JPR, JR, SR, JPR, RMR, IR, OS, RRS, PS, SS, BS, AS, JSN, ES, JS, NS, LS, BS, OS, SKS, GS, SS, ES, MT, CTE, CTE, BT, RT, DTF, MT, MT, OMU, ICU, JVN, CV, SV, IV, AVB, IW, IW, MW, TW, MW, FW, ADW, ZX, JX, EZP, AZ, RAZ

Writing-orig. draft: VC

Writing – review & editing: VC, NM, SB, JB, CB, MJ, EWM, SM, NO, MS, SVDL, NIAA, SA, NAS, BA, IA, EA, AA, MA, IMA, MA, MA, RMA, RA, TA, PA, DA, FA, AB, RB, KB, EB, CB, AYB, RB, OB, MB, AB, KB, AB, OB, PCA, FC, ACV, TC, RKC, SC, GC, SDP, RD, SD, CDC, LDS, KD, SD, KMD, CD, DD, MD, UKHE, ME, BE, TWE, MF, AFB, ZF, XF, CF, CF, MF, SF, HF, JF, MF, SF, AG, PGV, MGV, WG, OG, OG, TG, JG, EG, MG, CGB, HG, DG, GMG, LG, HH, LNH, PH, ACHM, AH, GH, MH, MH, NI, MI, MTI, YJ, TJ, CAJ, SJ, DJ, MZK, JK, SK, JRK, MK, TKR, OK, HK, AK, LK, EK, LK, LK, AK, JK, LSK, PK, SK, TK, AK, EAK, CL, AL, AL, OLC, JBL, ZL, NL, AL, GL, AL, ALO, CLV, NML, CHL, KLT, MDM, SM, RM, HM, JM, TLM, JM, PM, FMR, MM, IM, ZM, JN, EJN, JPN, FLNC, NNVN, DN, TO, TO, JPH, MP, PP, PPS, MPC, MP, YP, ARP, MP, CRP, KP, MP, JP, DP, AP, KP, EP, KP, DMQ, PR, AR, FGR, CRS, GR, JPR, JR, SR, JPR, RMR, IR, OS, RRS, PS, SS, BS, AS, JSN, ES, JS, NS, LS, BS, OS, SKS, GS, SS, BS, ES, MT, CTE, CTE, BT, AKT, RT, DTF, MT, MT, OMU, ICU, JVN, CV, SV, IV, MV, AVB, IW, IW, MW, TW, MW, FW, ADW, ZX, JX, EZP, AZ, RAZ

Competing interests: The authors declare no competing interests.

Supplementary Information is available for this paper.

Affiliations:

¹Department of the History of Science, Harvard University; Cambridge, 02138, USA.

²Department of Communication and Media Research, University of Zurich; Zurich, 8050, Switzerland.

³Institute of Sociology, University Bern; Bern, 3012, Switzerland.

⁴Department of Advertising + Public Relations, Michigan State University; East Lansing, 48824, USA.

⁵Department of Psychology, University of Amsterdam; Amsterdam, 1018 WT, The Netherlands.

⁶Department of Psychology, Inland Norway University of Applied Sciences; Elverum, 2418 Norway

⁷Centre for Research on Evaluation, Science and Technology, Stellenbosch University; Stellenbosch, 7600, South Africa.

⁸Centre for Climate Change Communication, George Mason University; Fairfax, 22030, USA.

⁹Centre for Research in Communication and Culture, Loughborough University; Loughborough, LE11, UK.

¹⁰Department of Psychology, University of Cambridge; Cambridge, CB2 3EB, UK.

¹¹Institute of Malaysian and International Studies, National University of Malaysia; Bangi, 43600, Malaysia.

¹²School of Collective Intelligence, Mohammed VI Polytechnic University; Ben Guerir, 43150, Morocco.

- ¹³Department of Science and Technology Studies, Faculty of Science, Universiti Malaya; Kuala Lumpur, 50603, Malaysia.
- ¹⁴ELTE Institute of Psychology, Eotvos Lorand University; Budapest, 1064, Hungary.
- ¹⁵Department of Psychology, University of Sheffield; Sheffield, S10 2TN, UK.
- ¹⁶Faculty of Psychology, Atma Jaya Catholic University of Indonesia; Jakarta, 12930, Indonesia.
- ¹⁷Department of Economics, University of Bath; Bath, BA2 7AY UK.
- ¹⁸Department of Philosophy, Macquarie University; Sydney, 2109, Australia.
- ¹⁹Department of Biochemistry, Faculty of Science, University of Dschang; Dschang, BP 96, Cameroon.
- ²⁰Sloan School of Management, Massachusetts Institute of Technology; Cambridge, 02142, USA.
- ²¹Department of Psychology, LMU Munich; Munich, 80802, Germany.
- ²²Division of Humanities and Social Science, California Institute of Technology; Pasadena, 91125, USA.
- ²³Department of Communication, George Mason University; Fairfax, 22030, USA.
- ²⁴Department of Physics, Egerton University; Egerton, 20115, Kenya.
- ²⁵Department of Psychology, Universitas Islam Negeri Sunan Kalijaga; Yogyakarta, 55281, Indonesia.
- ²⁶Department of Social Psychology, University of Groningen; Groningen, 9712, The Netherlands.
- ²⁷Museum of Natural Sciences “Sabiha Kasimati”, University of Tirana; Tirana, 1010, Albania.
- ²⁸Department of Architecture, University of Cambridge; Cambridge, CB2 3EB, UK.
- ²⁹Department of Biomedical Sciences, University of Botswana; Gaborone, Botswana.
- ³⁰Institute of Political Science, University of St. Gallen; St. Gallen, 9000, Switzerland.
- ³¹Institute for Planetary Health Behaviour, University of Erfurt; Erfurt, 99089, Germany.
- ³²Department of Economics, University of Birmingham; Birmingham, B15 2TT, UK.
- ³³Institute for Data, Systems, and Society, Massachusetts Institute of Technology; Cambridge, 02142, USA.
- ³⁴Department of Psychology, SWPS University of Social Sciences and Humanities; Warsaw, 03-815, Poland.
- ³⁵Faculty of Psychology, University of Warsaw; Warsaw, 00-183, Poland.
- ³⁶School of Psychology, University of Birmingham; Birmingham, B15 2TT, UK.
- ³⁷Computer Science Department, Harvey Mudd College; Claremont, 91711, USA.
- ³⁸Department of Psychology, Nantes Université, LPPL, Nantes, 44000, France
- ³⁹Institute for Sociology, Slovak Academy of Sciences; Bratislava, 813 64, Slovakia.
- ⁴⁰Department of Scientific and Innovation Culture. Spanish Foundation for Science and Technology; Alcobendas, Madrid, 28100, Spain.
- ⁴¹Department of International and Political Sciences, University of Genoa; Genoa, 16125, Italy.
- ⁴²Institute for Multimedia and Interactive Systems, University of Lübeck; Lübeck, 23562, Germany.
- ⁴³Department of Health Law, Policy, and Management, Boston University School of Public Health; Boston, 02118, USA.
- ⁴⁴Department of Sociology, University of the Philippines Diliman; Quezon City, 1101, Philippines.
- ⁴⁵Department of Psychology, Erzurum Technical University; Erzurum, 25050, Türkiye.
- ⁴⁶Institute of Psychology, Jagiellonian University; Kraków, 31-007, Poland.
- ⁴⁷Department of Psychology, KU Leuven; Leuven, 3000, Belgium.
- ⁴⁸Cambridge Zero, University of Cambridge; Cambridge, CB2 3EB, UK.
- ⁴⁹Department of Psychology, Université Rennes 2; Rennes, 35000, France.
- ⁵⁰Department of Cognition, Emotion, and Methods in Psychology, University of Vienna; Vienna, 1010, Austria.
- ⁵¹Institute of General Practice and Family Medicine, University of Bonn, University Hospital Bonn; Bonn, 53127, Germany.
- ⁵²School of Psychology, University of Kent; Canterbury, CT2 7NP, UK.

- ⁵³Harding Center for Risk Literacy, University of Potsdam; Potsdam, 14482, Germany.
- ⁵⁴Center for Sociocultural Research, HSE University; Moscow, 101000, Russia.
- ⁵⁵Department of Labor and Social Policy, University of Lodz; Lodz, 90-136, Poland.
- ⁵⁶School of Psychological Science, University of Western Australia; Perth, 6009, Australia.
- ⁵⁷Department of Management, Aarhus University; Aarhus, 8000, Denmark.
- ⁵⁸Department of Economics, Harvard University; Cambridge, 02138, USA.
- ⁵⁹Department of Political Science & Annenberg School for Communication, University of Pennsylvania; Philadelphia, 19104, USA.
- ⁶⁰Department of Computer Science and Engineering, University of Notre Dame; Notre Dame, 46556, USA.
- ⁶¹School of Communication and Culture, Aarhus University; Aarhus, 8000, Denmark.
- ⁶²a2i Programme of ICT Division and UNDP Bangladesh; Dhaka, 1207, Bangladesh.
- ⁶³Department of Resource Economics and Environmental Sociology, University of Alberta; Edmonton, T6G 2R3, Canada.
- ⁶⁴Department of Political Science and International Relations, Carleton College; Northfield, 55057, USA.
- ⁶⁵Faculty of Management and Economics, Ruhr-University Bochum; Bochum, 44801, Germany.
- ⁶⁶LMU Munich School of Management, LMU Munich; Munich, 80539, Germany.
- ⁶⁷Leibniz Institut für Wissensmedien; Tübingen, 72076, Germany.
- ⁶⁸Department of Psychology, Eberhard Karls Universität Tübingen; Tübingen 72076, Germany.
- ⁶⁹School of Social Work, Lucerne University of Applied Sciences and Arts; Lucerne, 6002, Switzerland.
- ⁷⁰Department of Psychology, Saarland University; Saarbrücken, 66123, Germany.
- ⁷¹Department of Political Science, Aarhus University; Aarhus, 8000, Denmark.
- ⁷²Department of Psychology, Universidad de Concepción; Concepción, 4030000, Chile.
- ⁷³Faculty of Health Sciences, University of Bristol; Bristol, BS8 1QU, UK.
- ⁷⁴Institute for Management & Organization, Leuphana University; Lueneburg, 21335, Germany.
- ⁷⁵School of Psychology, University of New South Wales; Sydney, 2052, Australia.
- ⁷⁶Research Institute for International Management, University of St. Gallen; St. Gallen, 9000, Switzerland.
- ⁷⁷Department of Business Administration, Instituto Tecnológico Autónomo de México; Mexico City, 01080, Mexico.
- ⁷⁸School of Psychology, Victoria University of Wellington; Wellington, 6012, New Zealand.
- ⁷⁹Department of Media and Communication, LMU Munich; Munich, 80799, Germany.
- ⁸⁰Department of Geography, University of Bergen; Bergen, 5020, Norway.
- ⁸¹Faculty of Data and Decision Sciences, Technion - Israel Institute of Technology; Haifa, 3200003, Israel.
- ⁸²Hixon Center for Climate and the Environment, Harvey Mudd College; Claremont, 91711, USA.
- ⁸³Faculty of Management, University of Warsaw; Warsaw, 02-678, Poland.
- ⁸⁴Secretaría de Planeación, Centro de Investigación y de Estudios Avanzados del Instituto Politécnico Nacional; Mexico City, 07360, Mexico.
- ⁸⁵Department of Media and Communication, City University of Hong Kong; Hong Kong.
- ⁸⁶School of Education, Trinity College Dublin; Dublin, D02 PN40, Ireland.
- ⁸⁷Department of International Relations and Regional Studies, KIMEP University; Almaty, 050010, Kazakhstan.
- ⁸⁸Graduate School of Interdisciplinary Information Studies, University of Tokyo; Tokyo, 113-0033, Japan.
- ⁸⁹Department of Government & Politics, Jahangirnagar University; Dhaka, 1342, Bangladesh.
- ⁹⁰Department of Psychology, University of Minnesota; Minneapolis, 55455, USA.
- ⁹¹School of Geography, Planning, and Spatial Sciences, University of Tasmania; Tasmania, 7005, Australia.
- ⁹²Political Sociology, University of Bamberg; Bamberg, 96047, Germany.

- ⁹³Institute of Psychology, Nicolaus Copernicus University; Toruń, 87-100, Poland.
- ⁹⁴Graduate Institute of Journalism, National Taiwan University; Taipei, 106216, Taiwan.
- ⁹⁵Department of Public Health, University of Otago; Wellington, 6021, New Zealand.
- ⁹⁶Department of Journalism and Mass Communication, Lviv Polytechnic National University; Lviv, 79000, Ukraine.
- ⁹⁷Institute of Communication Studies and Journalism, Charles University; Prague, 116 36, Czech Republic.
- ⁹⁸Center for Social and Cultural Psychology, Université Libre de Bruxelles; Brussels, 1050, Belgium.
- ⁹⁹School of Sciences and Humanities, Nazarbayev University; Astana, 010000, Kazakhstan.
- ¹⁰⁰Department of Social Research, University of Turku; Turku, 20500, Finland.
- ¹⁰¹Faculty of Life Sciences: Food, Nutrition and Health, University of Bayreuth; Kulmbach, 95326, Germany.
- ¹⁰²Department of Management and Engineering, Linköping University; Linköping, 581 83, Sweden.
- ¹⁰³Department of Psychology, Ural Federal University; Yekaterinburg, 620075, Russia.
- ¹⁰⁴Department of Information Science and Media Studies, University of Bergen; Bergen, 5020, Norway.
- ¹⁰⁵Centre for Climate and Energy Transformation (CET), University of Bergen; Bergen, 5020, Norway.
- ¹⁰⁶Department of Communication Science and Political Science, Vrije Universiteit Amsterdam; Amsterdam, 1081, The Netherlands.
- ¹⁰⁷Faculty of Behavioural and Social Sciences, University of Groningen; Groningen, 9712, The Netherlands.
- ¹⁰⁸Department of Communication and Internet Studies, Cyprus University of Technology; Limassol, 3036, Cyprus.
- ¹⁰⁹Laboratoire Parisien de Psychologie Sociale, Université Paris Nanterre; Nanterre, 92000, France.
- ¹¹⁰Laboratory for Research of Individual Differences, University of Belgrade; Belgrade, 11000, Serbia.
- ¹¹¹Department of Social Psychology and Methodology, Universidad Autónoma de Madrid; Madrid, 28049, Spain.
- ¹¹²School of Medicine and Psychology, Australian National University; Canberra, 2601, Australia.
- ¹¹³Uehiro Centre for Practical Ethics, University of Oxford; Oxford, OX1 1PT, UK.
- ¹¹⁴Institut Langage et Communication, University of Louvain; Louvain-la-Neuve, 1348, Belgium.
- ¹¹⁵Departamento de Psicología, Universidad Peruana Cayetano Heredia; La Molina, 15023, Peru.
- ¹¹⁶Department of Psychology, University of Victoria; Victoria, V8P 5C2, Canada.
- ¹¹⁷Harvard Kennedy School's Shorenstein Center, Harvard University; Cambridge, 02138, USA.
- ¹¹⁸Network Science Institute, Northeastern University; Boston, 02115, USA.
- ¹¹⁹School of Psychology and Public Health, La Trobe University; Melbourne, 3086, Australia.
- ¹²⁰Department of Psychology, Royal Holloway, University of London; Egham, TW20 0EX, UK.
- ¹²¹Institut Jean Nicod, Département d'études cognitives, ENS, EHESS, PSL University, CNRS; Paris, 75005, France.
- ¹²²Department of Communication Science, University of Muenster; Münster, 48143, Germany.
- ¹²³School of Psychology, University of Waikato; Hamilton, 3240, New Zealand.
- ¹²⁴Department of Political Science and International Relations, University of Delaware; Newark, 19716, USA.
- ¹²⁵Office for Quality Assurance, Analyses and Reporting, Project EUTOPIA, University of Ljubljana; Ljubljana, 1000, Slovenia.
- ¹²⁶Department of Management and Supply Chain Studies, Nkumba University; Entebbe, 0256, Uganda.
- ¹²⁷Department of User Experience, Intel; São Paulo, 05508-070, Brazil.
- ¹²⁸Malaria Epidemiology and Drug Resistance Laboratory, University of Buea; Buea, Cameroon.
- ¹²⁹Communication Arts Programme, Bowen University; Iwo, 284, Nigeria.
- ¹³⁰Department of Psychology and Psychotherapy, Witten/Herdecke University; Witten, 58455, Germany.
- ¹³¹Department of Management, University of Adger; Kristiansand, 4604, Norway.

- ¹³²Faculty of Humanities and Social Sciences, Trinity Western University; Langley, V2Y 1Y1, Canada.
- ¹³³Department of Clinical Neuroscience, Karolinska Institutet; Stockholm, 171 77, Sweden.
- ¹³⁴Institute of Psychology, University of Silesia in Katowice; Katowice, 40-007, Poland.
- ¹³⁵Penn Center for Neuroaesthetics, University of Pennsylvania; Philadelphia, 19104, USA.
- ¹³⁶Department of Psychological Science, Pomona College; Claremont, 91711, USA.
- ¹³⁷School of Psychology, Aston University; Birmingham, B4 7ET, UK.
- ¹³⁸Department of Psychology, University of Crete; Rethymno, 74100, Greece.
- ¹³⁹Science Studies Laboratory, University of Warsaw; Warsaw, 00-927, Poland.
- ¹⁴⁰Department of Psychology, University of the Philippines Diliman; Quezon City, 1101, Philippines.
- ¹⁴¹Max Planck Institute for Human Development; Berlin, 14195, Germany.
- ¹⁴²Social and Cognitive Neuroscience Laboratory, Mackenzie Presbyterian University; São Paulo, 01302-907, Brazil.
- ¹⁴³Department of Social Sciences, University of Hamburg; Hamburg, 20144, Germany.
- ¹⁴⁴Institut des Géosciences de l'Environnement, University Grenoble Alpes, CNRS, IRD, Grenoble-INP; Grenoble, 38000, France.
- ¹⁴⁵Institute of Environmental Health, Lisbon School of Medicine, University of Lisbon; Lisbon, 1649-028, Portugal.
- ¹⁴⁶Department of Implementation Research, Bernhard-Nocht-Institute for Tropical Medicine; Hamburg, 20359, Germany.
- ¹⁴⁷Centre for Language Studies, Radboud University Nijmegen; Nijmegen 6500 HD, The Netherlands
- ¹⁴⁸Department of Nutritional Sciences, University of Vienna; Vienna, 1090, Austria.
- ¹⁴⁹Department of Cognitive Psychology, Universität Hamburg; Hamburg, 20146, Germany.
- ¹⁵⁰School of Politics and International Relations, Australian National University; Canberra, 2601, Australia.
- ¹⁵¹Independent Researcher
- ¹⁵²Département de Sociologie, Université Officielle de Bukavu; Bukavu, 570, DR Congo..
- ¹⁵³Faculté des Sciences Sociales, Université Catholique de Bukavu; Bukavu, 285, DR Congo.
- ¹⁵⁴Department of Environmental Systems Science, ETH Zurich; Zurich, 8092, Switzerland.
- ¹⁵⁵Sociology Department, Vrije Universiteit Brussel; Brussel, 1050, Belgium.
- ¹⁵⁶UNSW Institute for Climate Risk & Response, University of New South Wales; Sydney, 2066, Australia.
- ¹⁵⁷Department of Psychology and Neuroscience, Boston College; Chestnut Hill, 02467, USA.
- ¹⁵⁸Faculty of Political Science and Economics, Waseda University; Tokyo, 169-8050, Japan.
- ¹⁵⁹Department of Law, University "Aleksandër Moisiu" Durrës; Durrës, 2009, Albania.
- ¹⁶⁰Centre for the Politics of Feelings, University of London; London, WC1E 7HU, UK.
- ¹⁶¹Division of Public Policy, The Hong Kong University of Science and Technology; Hong Kong
- ¹⁶²School of Psychology, University of Sussex; Falmer, BN1 9QH, UK.
- ¹⁶³Molecular Haematology and Immunogenetics Laboratory, Department of Medical Laboratory Science, Faculty of Health Sciences and Technology, College of Medicine, University of Nigeria Nsukka; Nsukka, 410105, Nigeria.
- ¹⁶⁴School of Sciences, UCLan Cyprus; Pyla, 7080, Cyprus
- ¹⁶⁵Department of Psychology, Education and Child Studies, Erasmus University Rotterdam; Rotterdam, 3062, The Netherlands.
- ¹⁶⁶Department of Psychology, New York University; New York, 10003, USA.
- ¹⁶⁷Department of Technology and Bionics, Rhine-Waal University; Kleve, 47533, Germany.
- ¹⁶⁸Melbourne Centre for Behaviour Change, University of Melbourne; Melbourne, 3010, Australia.
- ¹⁶⁹Faculty of Philosophy and Social Science, Nicolaus Copernicus University; Toruń, 87-100, Poland.

¹⁷⁰School of Economics and Management, Tongji University; Shanghai, 200092, China.

¹⁷¹School of Environment, Tsinghua University; Beijing, 100084, China.